Module 5A: Magnetism

Student Copy — Comprehensive Lecture & Tutorial Guide

PH7302: Condensed Matter Physics I

Department of Physics
National Institute of Technology Calicut

Academic Year 2026–2027

Abstract

This document is the complete, self-contained student edition of **Module 5A: Magnetism** for PH7302 Condensed Matter Physics I (Department of Physics, NIT Calicut). It develops the magnetic properties of solids from foundational definitions of magnetisation, magnetostatic units and the Bohr–van Leeuwen theorem, through atomic diamagnetism (Langevin/Larmor theory and the quantum diamagnetic Hamiltonian), localised paramagnetism (Hund's rules, Landé g-factor, orbital quenching, Brillouin function and Curie's law) and itinerant paramagnetism (Pauli spin susceptibility and Landau diamagnetism). The second half treats the exchange interaction (Heitler–London H_2 model, Heisenberg Hamiltonian, direct, superexchange and RKKY mechanisms) and collective ordering: Weiss molecular-field theory, the Curie–Weiss law, spontaneous magnetisation, and the two-sublattice Néel model for antiferromagnetism. Advanced extension topics (Van Vleck paramagnetism, adiabatic demagnetisation, the Stoner criterion, magnons, domain walls, hysteresis, and ferrimagnetism) are explicitly demarcated with **[Optional / Advanced Read]** tags so students can distinguish core exam syllabus material from advanced enrichment. Dielectric, ferroelectric, piezoelectric and optical properties of solids will be covered in companion guide **Module 5B**. The edition features an upfront assignment of 24 curated problems with final answers, 3 fully solved in-text numerical derivations, 20+ targeted learning boxes, four vector data figures, a master comparison table, a symbol glossary, and reproducible Python verification scripts.

How to Use This Student Edition

This student edition is designed to take you from first principles to exam-ready command of magnetism. The material is dense in algebra but light in intuition, so the edition leans heavily on physical pictures, worked numbers and explicit checkpoints. Work through it in the following way.

1. **Distinguish Core Syllabus from Optional Enrichment.** Sections, sub-topics and assignment problems tagged with **[Optional / Advanced Read]** contain enriching physical applications and derivations (e.g. Van Vleck paramagnetism, adiabatic demagnetisation, Stoner band splitting, spin-wave heat capacity, Bloch walls, and ferrimagnets). These topics go beyond the compulsory PH7302 semester exam syllabus and are provided for physical completeness and competitive exam preparation (such as CSIR-NET / GATE). Non-interested students can safely skip them without losing syllabus marks.

2. **Attempt the Upfront Assignment Continuously.** The section immediately after the Table of Contents contains the complete Module 5A problem set, organised into six thematic parts that map onto the chapters below. Each problem states exact parameters and ends with the expected final answer or a short hint in brackets (`[Ans: ...]`). Attempt each problem as soon as you finish the corresponding chapter — do not save them all for the end.

3. **Study the In-Text Worked Examples.** Three fully solved model derivations are embedded in the chapters at the points where the relevant technique is introduced (atomic hydrogen

diamagnetism, the exchange field of iron, and magnon heat capacity). Every one of them is written out in SI units with the unit conversions shown explicitly. Cover the solution, work it yourself, then compare.

4. **Engage with the Learning Boxes.** Distinct box types appear throughout and each serves a specific purpose:
    - **Conceptual Checkpoints**: quick tests of understanding that expose the gap between recognising a formula and understanding it.
    - **Study Takeaways**: the one-sentence physical statements worth committing to memory.
    - **Problem-Solving Strategies**: systematic calculation recipes for recurring exam question types.
    - **Common Misconceptions & Pitfalls**: the traps that cost marks — especially the SI/CGS unit traps, which dominate this module.
    - **Key Results**: boxed equations that you should be able to reproduce from memory.
5. **Read the Figures as Arguments.** The four generated figures are not decoration: each one isolates a piece of physics that is hard to see from the algebra alone. The captions tell you what to look for.
6. **Reproduce the Computations.** Appendix A contains the complete Python source for all four figures. Running and modifying these scripts converts the qualitative statements of the text into quantitative intuition — and it is the fastest way to check whether you have really understood the Brillouin function or the Curie–Weiss law.

1 Module Overview and Syllabus Alignment

The structure of this module and the syllabus learning outcomes that it addresses are summarised in Table 1.

Optional Enrichment / Beyond-Syllabus Read: Curriculum Scope: Module 5A versus Module 5B

In the official NIT Calicut PH7302 syllabus, Module 5 encompasses:

"Dielectric properties of insulators – Piezoelectric and Ferroelectric materials – Optical properties of solids – Magnetism-exchange interaction, diamagnetism, paramagnetism, ferromagnetism and antiferromagnetism."

This document constitutes **Module 5A**, dedicated entirely to the magnetic properties. The dielectric, ferroelectric, piezoelectric and optical properties will be issued in companion guide **Module 5B**. Within Module 5A, sections tagged **[Optional / Advanced Read]** extend beyond the minimum compulsory syllabus and may be omitted by students focusing solely on semester exam essentials.

Magnetism is the one branch of condensed matter physics in which the classical and quantum worlds give qualitatively opposite answers. The thread running through this module is that *charge* motion produces only a vanishingly small diamagnetic response (the Bohr–van Leeuwen theorem), whereas *spin* and *quantised orbital motion* produce the entire palette of magnetic behaviour, from the 10^{-6} diamagnetism of copper to the 10^{5} paramagnetism of a ferromagnet.

Chapter	Core Topic	Key Learning Outcomes & Syllabus Role
Ch. 1	Foundations, Units and the Bohr–van Leeuwen Theorem	[**Core**] Magnetisation, **B**/**H**/**M** in SI and CGS; susceptibility and molar susceptibility; proof that classical mechanics predicts zero magnetisation.
Ch. 2	Atomic Diamagnetism	[**Core**] Larmor precession and the Langevin classical result; quantum diamagnetic Hamiltonian; closed-shell atoms; noble-gas trends.
Ch. 3	Localised Paramagnetism	[**Core**] Landé g-factor; Hund's rules; Brillouin function; Curie law; orbital quenching in $3d$ vs. $4f$. *(Sec. 3.5 Van Vleck is an Optional Read).*
Ch. 4	Itinerant Paramagnetism	[**Core**] Pauli exclusion principle; Pauli spin susceptibility; Landau diamagnetism. *(Sec. 4.3 Adiabatic Demagnetisation is an Optional Read).*
Ch. 5	The Exchange Interaction	[**Core**] Failure of dipolar interaction; Heitler–London H_2 model; singlet/triplet splitting; Heisenberg Hamiltonian; exchange mechanisms.
Ch. 6	Ferromagnetism	[**Core**] Weiss molecular-field theory; Curie–Weiss law and T_C; spontaneous magnetisation and $\beta = 1/2$. *(Sec. 6.4 Stoner criterion is an Optional Read).*
Ch. 7	Magnons, Domains and Hysteresis	[**Optional / Extension**] Magnon dispersion; Bloch's $T^{3/2}$ law; magnetostatic energy; Bloch-wall width; hysteresis loops.
Ch. 8	Antiferromagnetism and Ferrimagnetism	[**Core AFM**] Two-sublattice Néel model; $T_N = \mu C$; $\chi = 2C/(T + T_N)$; $\chi_\parallel$ vs. $\chi_\perp$ and the T_N cusp. *(Sec. 8.3 Ferrimagnetism is an Optional Read).*
Ch. 9	Synthesis and Formula Sheet	[**Core Revision**] Master comparison of magnetic classes; consolidated formula sheet for exam revision.

Table 1: Structure, syllabus coverage and learning outcomes for Module 5A (Magnetism). Note: Companion topics in Module 5 (Dielectrics, Ferroelectrics, Piezoelectrics & Optical Properties) are covered in Module 5B.

2 Physical Constants, Units and Notation

2.1 Constants Used Throughout

Useful Numerical Benchmarks

- $1\ \text{eV} = 1.6022 \times 10^{-19}$ J; hence $1\ \text{meV} = 1.6022 \times 10^{-22}$ J.
- Thermal energy at room temperature: $k_B T = 0.02585$ eV at $T = 300$ K.
- At $T = 300$ K and $B = 1$ T: $\mu_B B / k_B T = (5.788 \times 10^{-5}\ \text{eV})/(0.02585\ \text{eV}) = 2.24 \times 10^{-3}$, so the dimensionless field is $x = g_J J \times 2.24 \times 10^{-3} \ll 1$ for any realistic ion. This is why the weak-field (Curie) limit is the normal laboratory situation, and why it takes a strong magnet or a very low temperature to reach saturation.
- Fermi temperature of typical metals: $T_F \sim 5 \times 10^4$–10^5 K, so conduction electrons are essentially degenerate at room temperature.
- Curie temperature of iron: $T_C = 1043$ K, i.e. $k_B T_C \approx 0.090$ eV.
- $1\ \text{T} = 10^4$ G and $\mu_0 = 4\pi \times 10^{-7}\ \text{T m/A}$.

Field and Susceptibility Conventions — Read This Before Chapter 6

Magnetism is the classic source of lost marks through unit confusion. Kittel and Ashcroft & Mermin write in Gaussian (CGS) units, while most modern papers and instruments use SI. This edition is

Quantity	Symbol	SI Value	CGS Value
Planck constant	h	6.6261×10^{-34} J s	6.6261×10^{-27} erg s
Reduced Planck constant	$\hbar$	1.0546×10^{-34} J s	1.0546×10^{-27} erg s
Electron mass	m_e	9.1094×10^{-31} kg	9.1094×10^{-28} g
Elementary charge	e	1.6022×10^{-19} C	4.8032×10^{-10} statC
Boltzmann constant	k_B	1.3807×10^{-23} J/K	1.3807×10^{-16} erg/K
Avogadro constant	N_A	6.0221×10^{23} mol^{-1}	—
Bohr magneton	$\mu_B = e\hbar/2m_e$	9.2740×10^{-24} J/T	9.2740×10^{-21} erg/G
Bohr magneton (in eV)	μ_B	5.7884×10^{-5} eV/T	—
Permeability of vacuum	μ_0	$4\pi \times 10^{-7}$ H/m	1 (Gaussian)
Electron gyromagnetic ratio	g	≈ 2.0023	≈ 2.0023
Bohr radius	a_0	5.2918×10^{-11} m	5.2918×10^{-9} cm

Table 2: Fundamental constants used in this module, in both unit systems.

explicit about which is in use at every step.

Quantity	SI	CGS (Gaussian)
Constitutive relation	$\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M})$	$\mathbf{B} = \mathbf{H} + 4\pi\mathbf{M}$
Volume susceptibility	$\chi = \partial M/\partial H \approx \mu_0 M/B$	$\chi = \partial M/\partial H \approx M/B$
Relation between the two	$\chi_{\text{SI}} = 4\pi\,\chi_{\text{CGS}}$	(same physical sample)
Molar susceptibility definition	$\chi_M^{\text{SI}} = \chi_{\text{SI}}\, V_m^{\text{SI}}$	$\chi_M^{\text{CGS}} = \chi_{\text{CGS}}\, V_m^{\text{CGS}}$
Molar conversion relation	$\chi_M^{\text{SI}} = 4\pi \times 10^{-6}\,\chi_M^{\text{CGS}}$	(χ_M^{SI} in m^3/mol, χ_M^{CGS} in cm^3/mol)
Bohr magneton	9.274×10^{-24} J/T	9.274×10^{-21} erg/G
Molar susceptibility unit	m^3/mol	cm^3/mol

Table 3: Field and susceptibility conventions in the two unit systems used in this module.

Common Misconception & Pitfall: The SI/CGS trap in this module

The SI and CGS magnetisation equations look almost identical but are *not*:

$$\text{SI:}\quad \chi = \frac{\mu_0 M}{B}, \qquad \text{CGS:}\quad \chi = \frac{M}{B}.$$

In SI the permeability μ_0 appears in χ, in CGS a factor 4π appears in $\mathbf{B} = \mathbf{H} + 4\pi\mathbf{M}$, and the dimensionless volume susceptibilities satisfy $\chi_{\text{SI}} = 4\pi\,\chi_{\text{CGS}}$.

Molar susceptibility conversion. Because volume susceptibility contains that factor of 4π, the molar susceptibility conversion also involves 4π:

$$\chi_M^{\text{SI}} = \chi_{\text{SI}}\, V_m^{\text{SI}} = (4\pi\,\chi_{\text{CGS}})(10^{-6}\, V_m^{\text{CGS}}) = 4\pi \times 10^{-6}\,\chi_M^{\text{CGS}}.$$

A classic examination error is simply multiplying χ_M^{CGS} by 10^{-6} while omitting the 4π, which leaves the SI molar susceptibility an order of magnitude too small!

Convention adopted in this edition. Chapters 1–5 use SI. From Chapter 6 onwards the mean-field (Weiss, Néel, ferrimagnet) derivations are carried out in **CGS**, following Kittel, because it keeps the algebra clean:

$$M = \frac{C}{T}\,(H_a + \lambda M) \quad\Longrightarrow\quad \chi = \frac{C}{T - T_C}, \qquad T_C = \lambda C = \frac{2zJS(S+1)}{3k_B}.$$

Whenever an SI form is needed it is given alongside. Every answer in the assignment set states its unit system.

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Assignment: Module 5A — Magnetism

Instructions to Students. This assignment covers the derivations, physical concepts and quantitative techniques developed across Module 5A. The problems are grouped into six thematic parts that follow the chapter order of this guide. Problems marked with [**Optional / Advanced Read**] address extension topics and may be skipped by students focusing only on compulsory semester exam questions. Each problem supplies exact parameters and closes with the expected final answer or an essential hint in square brackets (`[Ans: ...]`). Full step-by-step model derivations for the recurring problem archetypes appear as *Worked Examples* inside the corresponding chapters — attempt each problem before turning to them. State the unit system (SI or CGS) in every answer. Unless told otherwise, use $m_e = 9.1094 \times 10^{-31}$ kg, $e = 1.6022 \times 10^{-19}$ C, $\mu_0 = 4\pi \times 10^{-7}$ H/m and $k_B = 1.3807 \times 10^{-23}$ J/K.

Part I: Foundations, Units and Atomic Diamagnetism

I.1. Converting between CGS and SI susceptibilities.
A solid specimen has a Gaussian volume susceptibility $\chi_{\text{CGS}} = -4.5 \times 10^{-6}$ and a density $\rho = 9.2\ \text{g cm}^{-3}$ with molar mass $M_{\text{mol}} = 184.7\ \text{g mol}^{-1}$.

(a) Convert to the SI volume susceptibility.

(b) Compute the molar susceptibility χ_M in cm^3/mol.

(c) Compute the molar susceptibility in SI units, m^3/mol.

[**Ans:** (a) $\chi_{\text{SI}} = 4\pi\chi_{\text{CGS}} = -5.65 \times 10^{-5}$; (b) $V_m = M_{\text{mol}}/\rho = 20.08\ \text{cm}^3/\text{mol}$, so $\chi_M^{\text{CGS}} = \chi_{\text{CGS}} V_m = -9.04 \times 10^{-5}\ \text{cm}^3/\text{mol}$; (c) $\chi_M^{\text{SI}} = \chi_{\text{SI}} V_m^{\text{SI}} = 4\pi \times 10^{-6} \chi_M^{\text{CGS}} = -1.135 \times 10^{-9}\ \text{m}^3/\text{mol}$ (remember the factor of 4π must be included when converting to SI)]

I.2. Langevin diamagnetic susceptibility of a many-electron atom.
An atom carries $Z = 3$ bound electrons with mean-square radius $\langle r^2 \rangle = 0.45\ \text{Å}^2$, and the sample has $N = 6.0 \times 10^{28}$ atoms per cubic metre. Compute the volume diamagnetic susceptibility in SI. [**Ans:** $\chi_{\text{dia}} = -\mu_0 N Z e^2 \langle r^2 \rangle / 6 m_e = -4.78 \times 10^{-6}$ (dimensionless, SI)]

I.3. Atomic hydrogen: complete the Bohr calculation.
Using $\psi_{1s} = (\pi a_0^3)^{-1/2} e^{-r/a_0}$ with $a_0 = 0.529$ Å, show that $\langle r^2 \rangle = 3a_0^2$ and calculate the molar diamagnetic susceptibility of atomic hydrogen. [**Ans:** $\langle r^2 \rangle = 3a_0^2 = 8.40 \times 10^{-17}\ \text{cm}^2$; $\chi_M = -\mu_0 N_A e^2 a_0^2 / 2 m_e = -2.98 \times 10^{-11}\ \text{m}^3/\text{mol} = -2.37 \times 10^{-6}\ \text{cm}^3/\text{mol}$ (CGS molar)]

I.4. Larmor precession and the induced moment.
An electron with mean-square transverse radius $\langle \rho^2 \rangle = 0.30\ \text{Å}^2$ is placed in a perpendicular field $B = 2.0$ T. Compute (a) the induced magnetic moment per atom and (b) the induced current. [**Ans:** (a) $\mu_{\text{ind}} = -\frac{Ze^2 B}{4m_e} \langle \rho^2 \rangle = -4.23 \times 10^{-29}$ J/T; (b) with $A = \pi \langle \rho^2 \rangle = 9.42 \times 10^{-21}\ \text{m}^2$, $I = \mu / A = -4.48 \times 10^{-9}$ A (equivalently $I = -Ze^2 B / 4\pi m_e$)]

I.5. The Bohr–van Leeuwen theorem, stated and defended.
(a) Write the classical canonical partition function for N particles of charge q_i in a vector potential $\mathbf{A}(\mathbf{r})$ with $\mathbf{B} = \nabla \times \mathbf{A}$. (b) Show explicitly, by shifting the integration variable, that Z does not depend on $\mathbf{B}$. (c) Explain in one sentence what physical cancellation the algebra encodes. [**Ans:** (a) $Z = \frac{1}{N! h^{3N}} \int d^{3N}r\, d^{3N}p\, \exp[-\frac{\beta}{2m} \sum_i (\mathbf{p}_i - q_i \mathbf{A}(\mathbf{r}_i))^2 - \beta U]$; (b) substitute $\mathbf{p}_i' = \mathbf{p}_i - q_i \mathbf{A}(\mathbf{r}_i)$: the limits $(-\infty, +\infty)^3$ are unchanged and $\mathbf{A}$ drops out identically, so $Z(\mathbf{B}) = Z(0)$, F is $\mathbf{B}$-independent and $\mathbf{M} = -\frac{1}{V} \partial F / \partial \mathbf{B} = 0$; (c) the

boundary ("skipped") orbits that would carry the bulk current exactly cancel the bulk orbital currents of the interior atoms]

I.6. Noble-gas trend.
Explain, in terms of $\langle r^2 \rangle$ and Z, why the diamagnetic molar susceptibility of the noble gases grows along the series He, Ne, Ar, Kr, Xe, and predict the sign of the trend for the halogen family. [**Ans:** $\chi_{\text{dia}} \propto -NZ\langle r^2 \rangle$ and both Z and the shell radius grow down the group, so $|\chi_M|$ increases monotonically (He -1.9, Ne -7.2, Ar -19.4, Kr -28.0, Xe -43.0, all in 10^{-6} cm^3/mol); the same argument applies to the halogens, so $|\chi_M|$ should also grow downwards in the group]

Part II: Localised Paramagnetism, Hund's Rules and the Brillouin Function

II.1. Hund's rules for Fe^{2+} and Fe^{3+}.
For Fe^{2+} ($3d^6$) and Fe^{3+} ($3d^5$), determine the ground-state term $^{2S+1}L_J$, the Landé g-factor, the unquenched effective magneton number $g_J\sqrt{J(J+1)}$, and the quenched value $2\sqrt{S(S+1)}$. [**Ans:** Fe^{3+}: $S = 5/2$, $L = 0$, $J = 5/2$, $^6S_{5/2}$, $g_J = 2$, both formulas give $p_{\text{eff}} = \sqrt{35} = 5.92$ ($p_{\text{exp}} = 5.9$, $L = 0$ so nothing to quench). Fe^{2+}: $S = 2$, $L = 2$, shell more than half full so $J = L + S = 4$, $^5D_4$, $g_J = 3/2$, unquenched $p_{\text{eff}} = 1.5\sqrt{20} = 6.71$, quenched $p_{\text{eff}} = 2\sqrt{6} = 4.90$ ($p_{\text{exp}} = 5.4$, in between — confirms partial orbital quenching)]

II.2. Co^{2+}: a test of orbital quenching.
Co^{2+} is $3d^7$. Find the free-ion term, g_J, the free-ion p_{eff}, and the quenched p_{eff}; compare with the measured value ≈ 3.7 for Co^{2+} in CoO. [**Ans:** $S = 3/2$, $L = 3$ (F), more than half full so $J = L + S = 9/2$, term $^4F_{9/2}$; $g_J = 1 + \frac{24.75+3.75-12}{49.5} = 4/3$; $p_{\text{eff}}^{\text{free}} = \frac{4}{3}\sqrt{24.75} = 6.63$; quenched $p_{\text{eff}} = 2\sqrt{3.75} = 3.87$; the measured 3.7 is close to the quenched value, showing that the $3d$ orbital moment is quenched in the oxide host]

II.3. Extracting a Curie constant and a Weiss temperature from two measurements.
A paramagnetic salt gives $\chi = 2.4 \times 10^{-3}$ at $T = 300$ K and $\chi = 1.6 \times 10^{-3}$ at $T = 400$ K (SI, volume susceptibility). Determine the Curie constant C and the Weiss temperature θ in $\chi = C/(T - \theta)$. [**Ans:** $C = 2.4 \times 10^{-3}(300 - \theta) = 1.6 \times 10^{-3}(400 - \theta)$ gives $\theta = +100$ K; then $C = 2.4 \times 10^{-3} \times 200 = 0.48$ K]

II.4. Number of unpaired electrons from the Curie constant.
A paramagnetic salt has a molar Curie constant $C = 0.375$ cm^3 K mol^{-1} (CGS) and one magnetic ion per formula unit. Use $\mu_{\text{eff}} = 2.828\sqrt{C}$ to determine the number of unpaired electrons. [**Ans:** $\mu_{\text{eff}} = 2.828\sqrt{0.375} = 1.732\ \mu_B$; comparing with $\mu_{\text{eff}} = \sqrt{n(n+2)}\,\mu_B$ gives $n(n+2) = 3$, i.e. $n = 1$ unpaired electron]

II.5. Evaluating the Brillouin function exactly.
For $J = 3/2$ and $x = 1.0$, (a) evaluate $B_J(x)$ from the closed form, (b) evaluate the weak-field approximation $\frac{J+1}{3J}x$, and (c) comment on the size of the error. [**Ans:** (a) $B_{3/2}(1) = \frac{4}{3}\coth\frac{4}{3} - \frac{1}{3}\coth\frac{1}{3} = 0.4957$; (b) $\frac{5/2}{9/2} \times 1 = 0.5556$; (c) the linear approximation overestimates by 12%, because at $x = 1$ the series is no longer in its asymptotic regime — an important warning against extrapolating the Curie law beyond $x \ll 1$]

II.6. Van Vleck paramagnetism **[Optional / Advanced Read]**
An ion with a $J = 0$ ground state has one excited state at gap $\Delta = 0.10$ eV with

$|\langle s|\mu_z|0\rangle| = 1.0\ \mu_B$. For a concentration $N = 1.0 \times 10^{28}$ m^{-3}, compute the Van Vleck susceptibility in SI and comment on its temperature dependence. [**Ans:** $\chi_{\rm VV} = 2\mu_0 N|\mu_z|^2/\Delta = 1.35 \times 10^{-4}$; it is *temperature independent* to leading order, because the excited states are far above k_BT and their thermal population contributes only a small, exponentially suppressed correction]

II.7. Quenching: when does it happen?
Give the two competing energy scales that decide whether the orbital moment of a transition-metal ion survives in a solid, state their typical magnitudes for $3d$ and $4f$ ions, and explain why $4f$ ions in rare-earth salts obey $p_{\rm eff} = g_J\sqrt{J(J+1)}$ so accurately. [**Ans:** the crystal-field energy $\mathcal{H}_{\rm CF}$ competes with the spin–orbit energy $\lambda\,\mathbf{L}\cdot\mathbf{S}$. For $3d$ ions $\mathcal{H}_{\rm CF} \sim$ 1–2 eV $\gg \lambda\,\mathbf{L}\cdot\mathbf{S} \sim$ 0.01–0.05 eV, so L–S coupling is destroyed and $\langle L_z\rangle = 0$; for $4f$ ions $\mathcal{H}_{\rm CF}$ is tiny because the $4f$ shell is buried under filled $5s^25p^6$ shells, so J remains a good quantum number and the free-ion formula holds. The exceptions are Sm^{3+} and Eu^{3+}, where low-lying J-multiplets are thermally populated at 300 K]

Part III: Conduction-Electron Paramagnetism and Adiabatic Demagnetisation

III.1. Pauli susceptibility of sodium.
For sodium, $n = 2.54\times10^{28}$ m^{-3} conduction electrons and $T_F = 3.7\times10^4$ K. Compute the Pauli spin susceptibility in SI and confirm that it is temperature independent to leading order. [**Ans:** $\chi_{\rm Pauli} = \frac{3\mu_0 n\mu_B^2}{2k_BT_F} = 8.06\times10^{-6}$ (SI); the expression contains T_F, not T, so χ is essentially constant in T — the hallmark of Pauli paramagnetism]

III.2. Landau diamagnetism and the net susceptibility of a metal.
(a) Show that the Landau diamagnetic contribution satisfies $\chi_{\rm Landau} = -\frac{1}{3}(m_e/m^*)^2\chi_{\rm Pauli}$. (b) For free electrons ($m^* = m_e$) what fraction of the Pauli susceptibility survives? (c) Recompute (b) for $m^* = 1.2\,m_e$. [**Ans:** (b) $\chi_{\rm total} = \frac{2}{3}\chi_{\rm Pauli}$; (c) $\chi_{\rm Landau} = -\frac{1}{3(1.2)^2}\chi_{\rm Pauli} = -0.2315\,\chi_{\rm Pauli}$, so $\chi_{\rm total} = 0.769\,\chi_{\rm Pauli}$]

III.3. Adiabatic demagnetisation of a paramagnetic salt **[Optional / Advanced Read]**
A salt is magnetised isothermally at $T_i = 1.0$ K in $B_i = 2.0$ T and then demagnetised adiabatically.

(a) What final temperature is reached if the internal dipolar field is $B_{\rm local} = 0.010$ T?

(b) What if $B_{\rm local} = 0.0020$ T?

(c) Why can the process not continue below the internal field?

[**Ans:** (a) $T_f = T_iB_{\rm local}/B_i = 0.0050$ K $= 5.0$ mK; (b) $T_f = 1.0$ mK; (c) the spins are no longer fully polarised when $B \sim B_{\rm local}$, so the isentropic assumption fails and the salt equilibrates with the internal field — this sets the millikelvin floor. Nuclear demagnetisation (e.g. Cu) reaches the microkelvin regime because nuclear spins have far smaller moments and hence much smaller internal fields]

III.4. Schottky anomaly of a two-level system.
For a spin-1/2 paramagnet with $E_\pm = \mp\mu_BB$, derive $U = -N\mu_BB\tanh(\mu_BB/k_BT)$ and $C_B = Nk_B\left(\frac{\mu_BB}{k_BT}\right)^2\mathrm{sech}^2\left(\frac{\mu_BB}{k_BT}\right)$. Locate the maximum and express $C_{\rm max}$ in units of Nk_B. [**Ans:** the maximum occurs where $x\tanh x = 1$, i.e. $x = 1.1997$ (so $\mu_BB/k_BT \approx 1.2$); $C_{\rm max} = Nk_Bx^2\mathrm{sech}^2x = 0.4392\,Nk_B$. Note the peak is a Schottky anomaly: it tends to zero as $1/T^2$ at high T]

Part IV: The Exchange Interaction and Ferromagnetism

IV.1. Why dipole coupling cannot order iron.
(a) Estimate the dipole–dipole interaction energy between two neighbouring Bohr magnetons in iron separated by $a = 2.0$ Å. (b) Express it as an equivalent temperature and compare with $T_C(\mathrm{Fe}) = 1043$ K. (c) State the ratio in orders of magnitude. [**Ans:** (a) $U_{\mathrm{dip}} = \frac{\mu_0 \mu_B^2}{4\pi a^3} = 1.08 \times 10^{-24}$ J $= 6.7 \times 10^{-6}$ eV; (b) $T_{\mathrm{dip}} = U_{\mathrm{dip}}/k_B = 0.078$ K; (c) $k_B T_C = 0.090$ eV, so the dipolar scale is smaller by a factor $\approx 1.3 \times 10^4$, i.e. about *four orders of magnitude*. The real ordering interaction must be *electric* in origin, mediated by the Pauli principle: the exchange interaction]

IV.2. Heitler–London: extracting J from the singlet and triplet energies.
For a two-electron system the singlet and triplet energies are $E_S = -6.00$ eV and $E_T = -4.20$ eV.

(a) Compute the exchange integral $J = (E_S - E_T)/2$.

(b) Deduce whether the coupling is ferromagnetic or antiferromagnetic, and justify the sign from the Pauli principle.

(c) Write the spin Hamiltonian that reproduces both energies.

[**Ans:** (a) $J = -0.90$ eV; (b) $J < 0$, so the *triplet* (parallel spins) lies higher and the singlet (antiparallel spins) is the ground state — antiferromagnetic coupling. Parallel-spin electrons must occupy a spatial state antisymmetric under exchange, which carries a node (a Fermi hole) between the electrons and lowers their Coulomb repulsion; the parallel-spin state is therefore penalised. (c) $\mathcal{H}_{\mathrm{eff}} = \mathrm{const} - 2J\,\mathbf{s}_1 \cdot \mathbf{s}_2$ with $J = (E_S - E_T)/2$ (CGS: $E_S - E_T = -1.80$ eV)]

IV.3. Exchange integral and molecular field of iron from T_C.
For bcc iron, $T_C = 1043$ K and $z = 8$. Assuming $S = 1/2$, (a) find the exchange integral J, (b) find the Weiss molecular field B_E at $T = 0$, and (c) find B_E for the more realistic assumption $S = 1$. [**Ans:** using $T_C = 2zJS(S+1)/3k_B$: (a) $J = \frac{3k_B T_C}{2zS(S+1)} = \frac{3 \times 89.9 \text{ meV}}{2 \times 8 \times 0.75} = 22.5$ meV; (b) $B_E = 2zJS/g\mu_B = 1.55 \times 10^3$ T; (c) for $S = 1$, $J = 8.43$ meV and $B_E = 1.16 \times 10^3$ T — about 10^3 T, i.e. roughly 10^4 times the strongest steady laboratory field available]

IV.4. The Stoner criterion applied to two metals **[Optional / Advanced Read]**
(a) Iron has $I = 1.0$ eV and $g(E_F) = 3.6$ states/(eV · atom). (b) Copper has $I = 0.60$ eV and $g(E_F) = 0.15$ states/(eV · atom). Decide in each case whether a spontaneous band splitting is expected, and explain the physical difference. [**Ans:** (a) $I\,g(E_F) = 3.6 > 1$ — ferromagnetic; (b) $I\,g(E_F) = 0.09 < 1$ — not ferromagnetic. In Fe, Co and Ni the Fermi level sits inside a high-density $3d$ band, so the exchange energy gain from splitting outweighs the kinetic-energy cost; in Cu the $3d$ band is filled and E_F lies in the low-density $4s$ band, so the cost dominates. (Pd and Pt are the interesting borderline cases: nearly ferromagnetic, Stoner-enhanced paramagnets)]

IV.5. Critical exponent from the mean-field equation.
For $S = 1/2$, the self-consistency equation is $m = \tanh(mT_C/T)$ with $m = M(T)/M_0$.

(a) Show that $m \approx \sqrt{3}\,(1 - T/T_C)^{1/2}$ just below T_C and identify the critical exponent β.

(b) Evaluate m at $T = 0.9\,T_C$ using (a), and then by solving the exact equation numerically.

(c) Why is the asymptotic form an over-estimate?

[**Ans:** (a) $\beta = 1/2$; (b) the asymptotic form gives $m = 1.732\sqrt{0.1} = 0.548$, while the exact numerical solution gives $m = 0.525$; (c) the series expansion retains only the leading term in $(1-T/T_C)$, and at $T/T_C = 0.9$ the neglected corrections ($\propto (1-T/T_C)$) are still a few percent, so the asymptotic form overestimates the true value by about 4% there (it is reliable only for $T \to T_C^-$)]

IV.6. Bethe–Slater curve.
For the direct exchange mechanism, describe qualitatively how the sign of J depends on the ratio r_a/r_{3d}, and give a physical reason for the change of sign. [**Ans:** small r_a/r_{3d} (Cr, α-Mn): the $3d$ orbitals overlap strongly, covalent bonding dominates and $J < 0$ (antiferromagnetic); intermediate r_a/r_{3d} (Fe, Co, Ni): the overlap is weak enough that the reduction of the intra-atomic Coulomb repulsion (the Fermi-hole effect) wins, giving $J > 0$ (ferromagnetic)]

Part V: Magnons, Domains and Hysteresis [Optional / Advanced Read]

V.1. Bloch-wall width in iron.
For bcc iron take $J = 2.0 \times 10^{-21}$ J (per nearest-neighbour link), $S = 1$, anisotropy constant $K = 4.8 \times 10^4$ J/m^3 and lattice constant $a = 2.87 \times 10^{-10}$ m. (a) Derive $\sigma_{\text{wall}} = \frac{\pi^2 JS^2}{Na^2} + KNa$ and the optimal N. (b) Compute the wall thickness $\delta = Na$. [**Ans:** (a) $N = \pi\sqrt{JS^2/(Ka^3)} = \pi\sqrt{2.0 \times 10^{-21}/(4.8 \times 10^4 \times 2.36 \times 10^{-29})} = 132$ atomic planes; (b) $\delta = 132 \times 2.87 \times 10^{-10} = 3.8 \times 10^{-8}$ m $= 38$ nm. Exchange alone favours an infinitely thick wall; the anisotropy energy of the intermediate spins is what stops it]

V.2. Magnon energies and the thermal wavevector.
For a ferromagnet with $J = 2.0 \times 10^{-21}$ J, $S = 1$ and $a = 2.87$ Å:

(a) Compute the magnon energy $\hbar\omega$ at the zone boundary $ka = \pi$.

(b) Compute the long-wavelength stiffness constant D in $\hbar\omega = Dk^2$.

(c) At $T = 10$ K, find the wavevector at which the magnon energy equals $k_B T$, and the corresponding wavelength.

[**Ans:** (a) $\hbar\omega = 4JS(1-\cos\pi) = 8JS = 1.60 \times 10^{-20}$ J $= 0.100$ eV; (b) $D = 2JSa^2 = 3.29 \times 10^{-40}$ J m^2; (c) $k_B T = 1.381 \times 10^{-22}$ J gives $k = \sqrt{k_B T/D} = 6.5 \times 10^8$ m^{-1}, so $\lambda = 2\pi/k = 9.7$ nm — the thermally populated magnons have wavelengths of a few tens of lattice constants, which is why the collective (rather than single-spin) description is the right one near T_C]

V.3. Bloch's $T^{3/2}$ law for iron.
Using $\Delta M/M(0) = \frac{0.0587}{SQ}\left(\frac{k_B T}{2JS}\right)^{3/2}$ with $S = 1$, $J = 2.1 \times 10^{-21}$ J and $Q = 2$ (bcc), compute $\Delta M/M(0)$ at $T = 100$ K and compare with experiment. [**Ans:** $\Delta M/M(0) = \frac{0.0587}{2}\left(\frac{1.381\times10^{-21}}{4.2\times10^{-21}}\right)^{3/2} = 5.5 \times 10^{-3}$, i.e. about 0.55% — in reasonable agreement with the measured magnetisation drop of iron near 100 K, which is one of the cleanest confirmations of the spin-wave picture]

V.4. Reading a hysteresis loop.
A measured M–H loop for a hard magnet gives $M_s = 1.7 \times 10^6$ A/m, $M_r = 1.4 \times 10^6$ A/m, $H_c = 5.0 \times 10^3$ A/m and a loop area of 2.5×10^3 J/m^3.

(a) Compute the remanence ratio M_r/M_s and the squareness of the loop.

(b) Compute the energy dissipated per cycle per unit volume.

(c) Convert H_c to tesla using $\mu_0 H_c = B_c$.

[**Ans:** (a) $M_r/M_s = 0.824$ — a very square, magnetically hard loop; (b) the loop area is the work done per cycle per unit volume, 2.5×10^3 J/m^3; (c) $\mu_0 H_c = 4\pi \times 10^{-7} \times 5.0 \times 10^3 = 6.28 \times 10^{-3}$ T $= 62.8$ G]

V.5. Why are domains formed at all?

In 200 words, explain why a large uniformly magnetised block of iron is not a strong permanent magnet, and what energy balance determines the domain-wall width. [**Ans:** a single-domain magnetised block carries a large stray field with magnetostatic energy $\int B^2/2\mu_0\, dV$; subdividing into Weiss domains with flux closure cancels most of the external field and lowers this energy, at the cost of creating domain walls. Each wall costs exchange energy (which wants the wall thick, $\propto 1/N$) and magnetocrystalline anisotropy energy (which wants it thin, $\propto N$); minimising the sum gives $N = \pi\sqrt{JS^2/(Ka^3)}$]

Part VI: Antiferromagnetism and Ferrimagnetism

VI.1. Néel temperature of manganese oxide.

For MnO the sublattice spin is $S = 5/2$ with $z = 6$ nearest neighbours and $T_N = 118$ K. Using $T_N = 2z|J|S(S+1)/3k_B$, find the exchange integral J. [**Ans:** $|J| = \frac{3k_B T_N}{2zS(S+1)} = \frac{3\times 1.3807\times 10^{-23}\times 118}{2\times 6\times 8.75} = 4.65 \times 10^{-23}$ J $= 0.29$ meV, antiferromagnetic ($J < 0$)]

VI.2. The three susceptibilities of an antiferromagnet.

An antiferromagnet has Curie constant $C = 0.40$ K and $T_N = 100$ K (SI). Compute $\chi(T_N)$, $\chi_\perp(0)$, $\chi_\parallel(0)$ and $\chi_{\text{poly}}(0)$. [**Ans:** $\chi(T_N) = 2C/(2T_N) = C/T_N = 4.0 \times 10^{-3}$; $\chi_\perp(0) = 2C/2T_N = 4.0 \times 10^{-3}$ (temperature independent in the ideal model); $\chi_\parallel(0) = 0$; $\chi_{\text{poly}}(0) = \frac{2}{3}\chi(T_N) = 2.67 \times 10^{-3}$]

VI.3. Interpreting $1/\chi$ for a ferromagnet and an antiferromagnet.

(a) What does a positive intercept on the T-axis of a $1/\chi$ vs. T plot indicate, and what is it called? (b) What does a negative intercept indicate? (c) Explain why the negative intercept does *not* mean the antiferromagnet orders at a negative temperature. [**Ans:** (a) ferromagnetic; the intercept is the Curie temperature T_C; (b) antiferromagnetic; the intercept is $-\Theta$ where $\Theta = |T_N|$ in the simplest model; (c) the intercept is an extrapolation of the *paramagnetic* branch beyond its domain of validity — the susceptibility is positive at every $T > 0$ and the material orders at $T_N > 0$; the negative sign simply records that the ordering is antiparallel. Only the linear region well above T_N should be used for the extrapolation]

VI.4. Spin-flop transition **[Optional / Advanced Read]**

For an antiferromagnet with exchange field $B_E = 10$ T and anisotropy field $B_A = 0.50$ T, (a) compute the spin-flop field $B_{\text{SF}} = \sqrt{2B_E B_A}$, and (b) explain physically what happens at B_{SF}. [**Ans:** (a) $B_{\text{SF}} = \sqrt{2 \times 10 \times 0.5} = \sqrt{10} = 3.16$ T; (b) below B_{SF} the sublattice moments stay along the easy axis ($\chi_\parallel \approx 0$); above it the antiparallel sublattice moments abruptly rotate to lie *perpendicular* to the field, where the susceptibility is $\chi_\perp > 0$, gaining Zeeman energy $-\frac{1}{2}\chi_\perp B_a^2$ that outweighs the anisotropy cost. The transition shows up as a peak in $\chi_\parallel(T)$ at T_N]

VI.5. The magnetic moment of magnetite **[Optional / Advanced Read]**

In one formula unit of Fe_3O_4 there are two Fe^{3+} ions ($3d^5$, $S = 5/2$, $L = 0$, 5 μ_B each) and one Fe^{2+} ion ($3d^6$, $S = 2$, L quenched, 4 μ_B). (a) What would the moment be if

all spins were parallel? (b) Explain why the measured saturation moment is 4.1 μ_B per formula unit. (c) What is the net moment of a Fe^{3+}/Fe^{2+} spinel with the inverse structure? [**Ans:** (a) $5+5+4=14$ μ_B if all parallel — six times too large; (b) in the inverse spinel one Fe^{3+} sits on the tetrahedral A site pointing down while the other Fe^{3+} and the Fe^{2+} sit on the octahedral B site pointing up: the two Fe^{3+} moments cancel exactly ($+5-5=0$) and only the Fe^{2+} moment survives, giving 4 μ_B, close to the measured 4.1 μ_B; (c) 4 μ_B per formula unit, directed along the B-site sublattice direction]

VI.6. Antiferromagnetic magnons and the heat capacity **[Optional / Advanced Read]**
Using the linear dispersion $\hbar\omega_k = 4|J|Sa|k|$ for an antiferromagnet in three dimensions, show that

$$C_{\text{mag}} = \frac{2\pi^2}{15}\, k_B \left(\frac{k_B T}{4|J|Sa}\right)^3 \propto T^3,$$

and contrast the exponent with the ferromagnetic case. [**Ans:** substituting $u = ck/k_BT$ with $c = 4|J|Sa$ into $U = \frac{1}{2\pi^2}\int_0^\infty k^2\, dk\, \frac{ck}{e^{ck/k_BT}-1}$ gives $U = \frac{\pi^2}{30c^3}(k_BT)^4$, hence $C = \frac{2\pi^2}{15}k_B\left(\frac{k_BT}{c}\right)^3 \propto T^3$. The contrast is instructive: a linear dispersion in d dimensions gives $C \propto T^{d+1}$, so a 3D antiferromagnet gives T^3 while a 3D ferromagnet (quadratic dispersion) gives $T^{3/2}$]

VI.7. Antiferromagnetic order detected by neutron diffraction.
In MnO the chemical FCC unit cell has $a = 4.43$ Å both above and below $T_N = 118$ K. X-ray diffraction sees no change, but neutron diffraction at 80 K shows extra peaks at half-integer positions $(1/2, 1/2, 1/2)$. Explain both observations. [**Ans:** X-rays scatter from the electron density, which is unchanged by a magnetic reordering — the chemical lattice is identical above and below T_N. Neutrons carry a magnetic moment and scatter additionally from the ordered moments, so the staggered (antiferromagnetic) arrangement doubles the magnetic unit cell to $2a = 8.85$ Å and produces superlattice reflections at $(1/2, 1/2, 1/2)$ — direct evidence of Néel order (Shull and Smart, 1949)]

3 Foundations: Definitions, Units and the Quantum Origin of Magnetism

3.1 The Magnetisation and the Susceptibility

Everything in this module follows from one thermodynamic definition and one constitutive relation.

Magnetisation

The magnetisation $\mathbf{M}$ is the magnetic dipole moment per unit volume:

$$\mathbf{M} = \frac{1}{V}\sum_i \boldsymbol{\mu}_i \tag{1}$$

It is fixed by the free energy of the system in an applied field:

$$\mathbf{M} = -\frac{1}{V}\left(\frac{\partial F}{\partial \mathbf{B}}\right)_{T,V} \tag{2}$$

with $F = -k_B T \ln Z$ the Helmholtz free energy. Equivalently, for a state of energy E_i, the induced moment is $\mu_i = -\partial E_i/\partial B$.

The sign in (2) is worth internalising: a system lowers its free energy by orienting its moments *along* the field, so the magnetisation points along $\mathbf{B}$.

The three fields you will meet are related by

$$\text{SI:}\quad \mathbf{B} = \mu_0(\mathbf{H}+\mathbf{M}), \qquad \text{CGS:}\quad \mathbf{B} = \mathbf{H} + 4\pi\mathbf{M} \tag{3}$$

where $\mathbf{H}$ is the field that free currents produce, $\mathbf{M}$ is the response of the material, and $\mathbf{B}$ is the field that charges actually feel. In SI, $\mathbf{H}$ and $\mathbf{M}$ have the *same* units (A/m); in CGS they differ by 4π.

The volume susceptibility is

$$\chi = \frac{\partial M}{\partial H} \approx \begin{cases} \dfrac{\mu_0 M}{B} & \text{(SI)}, \\ \dfrac{M}{B} & \text{(CGS)} \end{cases} \tag{4}$$

and the molar susceptibility, which is what tabulated data usually quote, is

$$\chi_M = \chi\, V_m = \chi\, \frac{M_{\text{mol}}}{\rho} \tag{5}$$

with V_m the molar volume. The two susceptibilities of the same sample satisfy $\chi_{\text{SI}} = 4\pi\chi_{\text{CGS}}$.

Study Takeaway: Three questions to ask of any magnetic material

1. **Is χ positive or negative?** (paramagnetic / diamagnetic)
2. **How does χ depend on T?** (the fingerprint of the mechanism)
3. **Is the response linear in the field?** (if not, the material is ordered)

3.2 Why Magnetism is Strictly Quantum: the Bohr–van Leeuwen Theorem

Conceptual Checkpoint: Why does a classical electron gas show no diamagnetism?

A magnetic field exerts a Lorentz force $\mathbf{F} = q(\mathbf{v} \times \mathbf{B})$ that curves electron trajectories into circles. If electrons really do orbit in circles, why does a classical gas of charged particles show *no* net diamagnetic moment in thermal equilibrium?
Hint: consider what a classical gas does at the *boundary* of the sample, not in the interior.

Proof. In classical statistical mechanics the canonical partition function for N particles of charge q_i in a vector potential $\mathbf{A}(\mathbf{r})$ (with $\mathbf{B} = \nabla \times \mathbf{A}$) is

$$Z = \frac{1}{N!\,h^{3N}} \int d^{3N}r \int d^{3N}p \, \exp\left[-\frac{\beta}{2m}\sum_{i=1}^{N}(\mathbf{p}_i - q_i\mathbf{A}(\mathbf{r}_i))^2 - \beta U(\mathbf{r}_1,\ldots,\mathbf{r}_N)\right] \tag{6}$$

For any fixed configuration $\{\mathbf{r}_i\}$ shift the momentum integration variable:

$$\mathbf{p}_i' = \mathbf{p}_i - q_i\mathbf{A}(\mathbf{r}_i) \quad \Longrightarrow \quad d^3p_i' = d^3p_i \tag{7}$$

Because the limits of integration run from $-\infty$ to $+\infty$, the shift is exact and $\mathbf{A}(\mathbf{r}_i)$ cancels identically from the integral. Hence

$$Z(\mathbf{B}) = Z(0) \implies F \text{ independent of } \mathbf{B} \implies \mathbf{M} = -\frac{1}{V}\frac{\partial F}{\partial \mathbf{B}} = \mathbf{0} \tag{8}$$

Study Takeaway: Bohr–van Leeuwen: the punchline

Classical mechanics predicts **exactly zero** magnetisation. The cancellation is physical, not algebraic bookkeeping: the bulk orbital currents of the interior atoms are exactly cancelled by the "skipped" orbits of the atoms at the boundary of the sample, which carry an equal and opposite current. Classical mechanics also predicts no spin, since spin is a quantum attribute. **Magnetism is therefore 100% a quantum phenomenon**, arising from quantised energy levels and intrinsic spin.

4 Atomic Diamagnetism: Larmor Precession and the Quantum Hamiltonian

Diamagnetism ($\chi < 0$) is the universal tendency of a bound electron to shield the interior of a body from an applied magnetic field — the microscopic analogue of Lenz's law. Unlike a free charge, an electron bound in a stationary atomic orbital maintains a persistent induced current for as long as $\mathbf{B}$ is applied.

4.1 Classical Langevin Derivation via Larmor Theorem

1. **Larmor precession.** By Larmor's theorem, the motion of electrons of charge $-e$ and mass m around a central nucleus in a weak field $\mathbf{B} = B\hat{z}$ is identical to the unperturbed motion plus a superimposed precession at the **Larmor angular frequency**

$$\omega_L = \frac{eB}{2m} \text{ (SI)}, \qquad \omega_L = \frac{eB}{2mc} \text{ (CGS)} \tag{9}$$

2. **Induced current.** For Z electrons per atom the precession constitutes an induced current loop:

$$I = (\text{charge}) \times (\text{revolutions per unit time}) = (-Ze)\frac{\omega_L}{2\pi} = -\frac{Ze^2B}{4\pi m} \tag{10}$$

3. **Induced moment.** For a loop of projected area $A = \pi\langle\rho^2\rangle = \pi\langle x^2 + y^2\rangle$,

$$\mu_{\text{ind}} = I \cdot A = -\frac{Ze^2B}{4m}\langle\rho^2\rangle \tag{11}$$

4. **Spherical symmetry.** For a spherically symmetric atom $\langle x^2\rangle = \langle y^2\rangle = \langle z^2\rangle = \frac{1}{3}\langle r^2\rangle$, so

$$\langle\rho^2\rangle = \langle x^2 + y^2\rangle = \frac{2}{3}\langle r^2\rangle \tag{12}$$

5. **Langevin susceptibility.** For N atoms per unit volume,

$$\chi_{\text{dia}} = \frac{\mu_0 N \mu_{\text{ind}}}{B} = -\frac{\mu_0 N Z e^2}{6m}\langle r^2\rangle \ \text{(SI)}, \qquad \chi_{\text{dia}} = -\frac{NZe^2}{6mc^2}\langle r^2\rangle \ \text{(CGS)} \tag{13}$$

Common Misconception & Pitfall: Diamagnetism is a small effect — do not expect a large answer

Because $\chi_{\text{dia}} \sim 10^{-6}$, several textbooks write $\chi_{\text{dia}} \approx -\frac{\mu_0 N Z e^2}{6m}\langle r^2\rangle$ and then quote a "molar susceptibility" in cm^3/mol. Remember that χ_M carries *volume* units: a value of -2.37×10^{-6} cm^3/mol for atomic hydrogen is a perfectly normal-sized diamagnetic answer, not a suspiciously small one.

4.2 Quantum Mechanical Derivation [Optional / Advanced Read]

In quantum mechanics we replace the canonical momentum $\mathbf{p} \to \mathbf{p} + e\mathbf{A}$ (for electron charge $q = -e$) and add the intrinsic spin Zeeman term. Choosing the symmetric gauge for a uniform field $\mathbf{B} = B\hat{z}$:

$$\mathbf{A} = \frac{1}{2}(\mathbf{B} \times \mathbf{r}) \implies A_x = -\frac{1}{2}yB, \quad A_y = \frac{1}{2}xB, \quad A_z = 0 \tag{14}$$

The kinetic energy of a sum over atomic electrons becomes

$$\mathcal{H}_{\text{kin}} = \sum_i \frac{1}{2m}\left(\mathbf{p}_i + e\mathbf{A}(\mathbf{r}_i)\right)^2 = \sum_i \frac{p_i^2}{2m} + \frac{e}{2m}\sum_i (\mathbf{p}_i \cdot \mathbf{A}_i + \mathbf{A}_i \cdot \mathbf{p}_i) + \frac{e^2}{2m}\sum_i A_i^2 \tag{15}$$

Since $\nabla \cdot \mathbf{A} = 0$, $\mathbf{p}$ and $\mathbf{A}$ commute, and using $\mathbf{r}_i \times \mathbf{p}_i = \hbar\mathbf{l}_i$:

$$\frac{e}{m}\sum_i \mathbf{A}_i \cdot \mathbf{p}_i = \frac{e}{2m}\sum_i (\mathbf{B} \times \mathbf{r}_i) \cdot \mathbf{p}_i = \frac{e}{2m}\mathbf{B} \cdot \sum_i (\mathbf{r}_i \times \mathbf{p}_i) = \frac{e\hbar}{2m}\mathbf{B} \cdot \mathbf{L} = \mu_B\, \mathbf{B} \cdot \mathbf{L} \tag{16}$$

where $\mu_B = e\hbar/2m = 9.274 \times 10^{-24}$ J/T (0.9274×10^{-20} erg/G) is the **Bohr magneton**, and $\mathbf{L}$ is the *dimensionless* orbital angular momentum operator ($\mathbf{L}_{\text{phys}} = \hbar\mathbf{L}$). Adding the spin term $g_0\mu_B\, \mathbf{B} \cdot \mathbf{S}$ with $g_0 \approx 2.0023 \approx 2$, and using $A_i^2 = \frac{1}{4}B^2(x_i^2 + y_i^2)$, we obtain

$$\boxed{\mathcal{H}_{\text{mag}} = \mu_B\,(\mathbf{L} + g_0\mathbf{S}) \cdot \mathbf{B} \ + \ \frac{e^2B^2}{8m}\sum_{i=1}^{Z}\left(x_i^2 + y_i^2\right)} \tag{17}$$

- **First term (paramagnetic, $\propto B$):** linear in B. First-order perturbation theory gives the permanent-dipole orientation and the Brillouin/Curie behaviour; second-order theory gives Van Vleck paramagnetism.
- **Second term (diamagnetic, $\propto B^2$):** quadratic in B. For a closed-shell atom ($\mathbf{L} = \mathbf{S} = \mathbf{J} = 0$) the first term vanishes identically, so this is the whole story. The first-order shift of the ground state $|0\rangle$ is

$$\Delta E_0 = \frac{e^2B^2}{8m}\sum_i \langle 0|x_i^2 + y_i^2|0\rangle = \frac{Ze^2B^2}{12m}\langle r^2\rangle \implies \mu = -\frac{\partial \Delta E_0}{\partial B} = -\frac{Ze^2\langle r^2\rangle}{6m}B \tag{18}$$

which recovers the Langevin result (13) exactly.

How to Read This Figure: Atomic hydrogen and the noble gases

For ground-state atomic hydrogen $\psi_{1s}(r) = (\pi a_0^3)^{-1/2} e^{-r/a_0}$, and one integrates $\langle r^2 \rangle = 3a_0^2$. For the noble gases the molar susceptibilities χ_M (in 10^{-6} cm^3/mol, CGS) grow monotonically with the atomic radius and Z: He (-1.9), Ne (-7.2), Ar (-19.4), Kr (-28.0), Xe (-43.0). Because $\chi_{\text{dia}} \propto -NZ\langle r^2 \rangle$, the trend is entirely geometric — which is why the halogens should show the same monotonic increase down their group.

Worked Example: Atomic hydrogen — molar diamagnetic susceptibility

$\psi_{1s} = (\pi a_0^3)^{-1/2} e^{-r/a_0}$ with $a_0 = 0.529$ Å, $Z = 1$.

Step 1 — the radial moment. The probability density is $|\psi|^2 = \frac{1}{\pi a_0^3} e^{-2r/a_0}$, so

$$\langle r^2 \rangle = \frac{4\pi}{\pi a_0^3} \int_0^\infty r^4 e^{-2r/a_0}\, dr = \frac{4}{a_0^3} \cdot \frac{4!}{(2/a_0)^5} = \frac{4 \cdot 24\, a_0^5}{32 a_0^3} \cdot \frac{1}{a_0^3}\Big|_{\text{simplify}} = 3a_0^2.$$

In SI, $a_0 = 5.2918 \times 10^{-11}$ m, so $\langle r^2 \rangle = 8.40 \times 10^{-21}$ m^2.

Step 2 — molar susceptibility. From (13), multiplying by the molar volume N_A/N,

$$\chi_M = -\frac{\mu_0 N_A e^2}{6m_e} \langle r^2 \rangle = -\frac{\mu_0 N_A e^2 a_0^2}{2m_e}.$$

Numerically: $\mu_0 N_A e^2 = (1.2566 \times 10^{-6})(6.0221 \times 10^{23})(2.5670 \times 10^{-38}) = 1.9432 \times 10^{-20}$, times $a_0^2 = 2.8003 \times 10^{-21}$ m^2 gives 5.4416×10^{-41}, divided by $2m_e = 1.8219 \times 10^{-30}$ kg:

$$\boxed{\chi_M = -2.99 \times 10^{-11} \text{ m}^3/\text{mol}}$$

which is -2.37×10^{-6} cm^3/mol in CGS molar units (the 4π factor is absorbed in the unit conversion).

Interpretation. The negative sign is the Lenz's-law response: the induced current opposes the applied field. The magnitude is set entirely by the size of the atom — which is why diamagnetism is temperature independent and essentially unaffected by anything you do to the electronic structure short of changing $\langle r^2 \rangle$.

5 Paramagnetism of Localised Moments: Hund's Rules and the Brillouin Function

Paramagnetism ($\chi > 0$) occurs in systems with unpaired electrons: odd-electron atoms and defects (F-centres, Na vapour), free radicals, and above all ions with partially filled inner shells ($3d$ transition metals, $4f$ lanthanides, $5f$ actinides).

5.1 Atomic Moments, the Landé g-Factor and Hund's Rules

For a free atom or ion in Russell–Saunders (L–S) coupling, $\mathbf{J} = \mathbf{L} + \mathbf{S}$. Because $\mathbf{L} + 2\mathbf{S}$ precesses rapidly about the conserved $\mathbf{J}$, we project the moment operator $\boldsymbol{\mu} = -\mu_B(\mathbf{L} + 2\mathbf{S})$ onto $\mathbf{J}$:

$$\boldsymbol{\mu}_{\text{eff}} = -g_J \mu_B \mathbf{J}, \qquad g_J = 1 + \frac{J(J+1) + S(S+1) - L(L+1)}{2J(J+1)} \tag{19}$$

with $g_J = 1$ for a pure orbital moment ($S = 0$) and $g_J = 2$ for a pure spin moment ($L = 0$).

Key Result: Hund's three rules for the ground-state multiplet $^{2S+1}L_J$

1. **Maximise S.** Electrons occupy separate orbitals with parallel spins, maximising the total spin S (subject to Pauli). This minimises the intra-atomic Coulomb repulsion because parallel-spin electrons are kept apart by the exchange/Fermi hole.
2. **Maximise L.** Subject to rule 1, maximise $L = \sum m_l$ by filling orbitals with the same m_l sense, so that electrons orbit in the same direction and stay farther apart.
3. **Fix J via spin–orbit coupling** ($\lambda\,\mathbf{L}\cdot\mathbf{S}$):
   - less than half-filled shell: $J = |L - S|$;
   - more than half-filled shell: $J = L + S$;
   - exactly half-filled shell (d^5, f^7): $L = 0 \Rightarrow J = S$ (an S-state ion).

5.2 Quantum Theory of Paramagnetism: the Brillouin Function

Consider N non-interacting magnetic ions per unit volume with total angular momentum J in a field $\mathbf{B} = B\hat{z}$. The $(2J+1)$-fold degenerate ground state splits into Zeeman levels indexed by $m_J \in \{-J, \ldots, +J\}$:

$$E(m_J) = -\mu_z B = g_J \mu_B m_J B \tag{20}$$

Defining the dimensionless ratio of magnetic to thermal energy

$$x \equiv \frac{g_J \mu_B J B}{k_B T} \implies E(m_J) = \frac{m_J}{J}\, x\, k_B T \tag{21}$$

Derivation. The single-ion canonical partition function Z_1 is a finite geometric series with common ratio $r = e^{-x/J}$:

$$Z_1 = \sum_{m_J=-J}^{+J} \exp\left(-\frac{m_J x}{J}\right) = e^{x} + e^{x(J-1)/J} + \cdots + e^{-x} \tag{22}$$

Summing the geometric series of $2J+1$ terms and multiplying numerator and denominator by $e^{-x/(2J)}$:

$$Z_1 = \frac{e^{x} - e^{-x(J+1)/J}}{1 - e^{-x/J}} = \frac{\sinh\!\left(\frac{2J+1}{2J}x\right)}{\sinh\!\left(\frac{1}{2J}x\right)} \tag{23}$$

The magnetisation $M = N\langle \mu_z \rangle$ follows from $F = -N k_B T \ln Z_1$:

$$M = -\frac{\partial F}{\partial B} = N k_B T \frac{\partial \ln Z_1}{\partial x}\frac{\partial x}{\partial B} = N g_J \mu_B J \frac{\partial}{\partial x}\left[\ln\sinh\!\left(\frac{2J+1}{2J}x\right) - \ln\sinh\!\left(\frac{1}{2J}x\right)\right] \tag{24}$$

Taking the derivative yields the celebrated

$$\boxed{M = N g_J \mu_B J\, B_J(x), \qquad x = \frac{g_J \mu_B J B}{k_B T}} \tag{25}$$

with the **Brillouin function**

$$\boxed{B_J(x) = \frac{2J+1}{2J}\coth\!\left(\frac{2J+1}{2J}x\right) - \frac{1}{2J}\coth\!\left(\frac{1}{2J}x\right)} \tag{26}$$

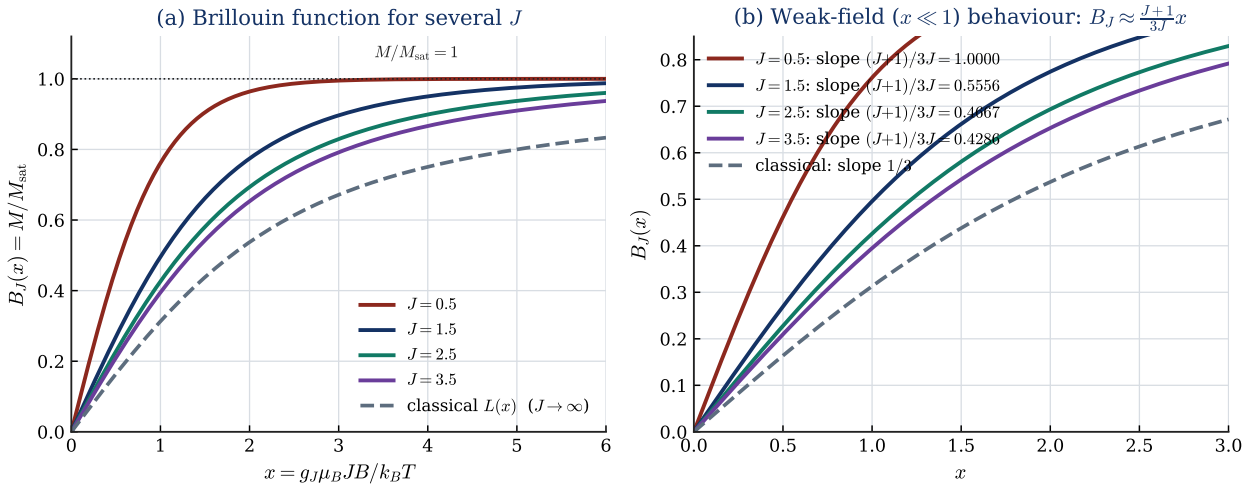


Figure 1: The Brillouin function. **(a)** $B_J(x)$ for several J together with the classical Langevin limit; all curves start at the origin with a finite slope and saturate at $B_J = 1$. **(b)** The weak-field regime, where every curve is a straight line of slope $(J+1)/(3J)$; the slope *decreases* with J, which is why larger J means a smaller Curie constant at fixed moment.

5.3 Limiting Cases of the Brillouin Function

1. **Two-level spin-1/2 system ($J = S = 1/2$, $L = 0$, $g_J = 2$).** Using $\coth(2x) - \frac{1}{2}\coth(x) = \tanh x$ (or directly $Z_1 = e^x + e^{-x} = 2\cosh x$):

$$B_{1/2}(x) = \tanh x \implies M = N\mu_B \tanh\left(\frac{\mu_B B}{k_B T}\right) \tag{27}$$

2. **Classical limit ($J \to \infty$).** When all orientations are continuously allowed, $\frac{2J+1}{2J} \to 1$ and $\frac{1}{2J}\coth\left(\frac{x}{2J}\right) \to \frac{1}{x}$:

$$\lim_{J\to\infty} B_J(x) = \coth x - \frac{1}{x} \equiv L(x) \tag{28}$$

3. **Saturation limit ($x \gg 1$).** Since $\coth u \to 1$ as $u \to \infty$, $B_J(x) \to \frac{2J+1}{2J} - \frac{1}{2J} = 1$, giving

$$M_{\rm sat} = N g_J \mu_B J \tag{29}$$

4. **Weak-field / high-temperature limit ($x \ll 1$).** At $T = 300$ K and $B = 1$ T, $\mu_B B/k_B T = (5.788\times10^{-5}\ \text{eV})/(0.02585\ \text{eV}) = 2.24\times10^{-3}$, so $x = g_J J \times 2.24\times10^{-3} \ll 1$ is the normal laboratory situation. Using $\coth u = \frac{1}{u} + \frac{u}{3} - \frac{u^3}{45} + \ldots$:

$$B_J(x) \approx \frac{2J+1}{2J}\left[\frac{2J}{(2J+1)x} + \frac{2J+1}{6J}x\right] - \frac{1}{2J}\left[\frac{2J}{x} + \frac{x}{6J}\right] = \frac{(2J+1)^2-1}{12J^2}x = \frac{J+1}{3J}x \tag{30}$$

Substituting $x = g_J\mu_B JB/k_BT$ into (25) gives **Curie's law**:

$$\boxed{\chi_{\rm Curie} = \frac{\mu_0 M}{B} = \frac{\mu_0 N g_J^2 \mu_B^2 J(J+1)}{3k_BT} = \frac{\mu_0 N p_{\rm eff}^2 \mu_B^2}{3k_BT} \equiv \frac{C}{T}} \tag{31}$$

where C is the **Curie constant** and $p_{\rm eff} = g_J\sqrt{J(J+1)}$ is the **effective Bohr magneton number**.

Study Takeaway: Reading the Brillouin figure

Look at **Figure 1(b)** again after this section. The slopes in the weak-field region are $(J+1)/(3J)$: for $J = 1/2$ it is 1.0, for $J = 3/2$ it is 0.556, and the classical limit is 1/3. A large J therefore means a *slower* approach to saturation per unit x but a *larger* saturation magnetisation $Ng_J\mu_B J$. These two effects compete, and the Curie constant $C \propto g_J^2 J(J+1)$ is what survives in the weak-field limit.

5.4 Rare Earth ($4f$) versus Iron-Group ($3d$): Orbital Quenching

Conceptual Checkpoint: When does the orbital moment survive in a solid?

Compare the energy scales $\mathcal{H}_{\text{CF}}$ (crystal electric field) and $\lambda\,\mathbf{L}\cdot\mathbf{S}$ (spin–orbit coupling) for a $3d$ ion and a $4f$ ion. Which one is larger in each case, and what does that do to p_{eff}?

- **Rare-earth ions ($4f^n$).** The $4f$ shell lies deep inside the ion, shielded by the filled $5s^2 5p^6$ outer shells. Crystal electric fields from neighbouring ions are weak compared with spin–orbit coupling ($\mathcal{H}_{\text{CF}} \ll \lambda\,\mathbf{L}\cdot\mathbf{S}$). Hence J remains a good quantum number and the experimental p_{eff} agrees excellently with $g_J\sqrt{J(J+1)}$ — with two exceptions: Sm^{3+} and Eu^{3+}, where low-lying excited J-multiplets are thermally populated at 300 K and p_{eff} falls below the free-ion value.
- **Iron-group ions ($3d^n$).** The $3d$ electrons sit on the outside of the ion and experience an intense, inhomogeneous crystal field ($\mathcal{H}_{\text{CF}} \sim$ 1–2 eV $\gg \lambda\,\mathbf{L}\cdot\mathbf{S} \sim$ 0.01–0.05 eV). The crystal field destroys the L–S coupling and splits the $(2L+1)$ orbital states into non-magnetic standing waves (e.g. real orbitals d_{xy}, $d_{x^2-y^2}$ in an octahedral or tetragonal field) for which $\langle L_z\rangle = 0$.
- **Result.** Only the spin contributes, so for $3d$ ions in solids

$$\boxed{p_{\text{eff}}(3d) \approx 2\sqrt{S(S+1)} = \sqrt{n(n+2)}} \tag{32}$$

with n the number of unpaired d electrons.

Table 4: Theoretical and experimental effective magneton numbers. Note how the $4f$ rows track the unquenched column while the $3d$ rows track the quenched one.

Ion (shell)	Ground term	$g_J\sqrt{J(J+1)}$	$2\sqrt{S(S+1)}$	p_{exp}	Regime
Nd^{3+} ($4f^3$)	$^4I_{9/2}$	3.62	3.87	3.5	Unquenched ($4f$)
Gd^{3+} ($4f^7$)	$^8S_{7/2}$	7.94	7.94	8.0	Half-filled
Dy^{3+} ($4f^9$)	$^6H_{15/2}$	10.63	5.92	10.6	Unquenched ($4f$)
Ti^{3+}, V^{4+} ($3d^1$)	$^2D_{3/2}$	1.55	1.73	1.8	Quenched ($3d$)
Cr^{3+} ($3d^3$)	$^4F_{3/2}$	0.77	3.87	3.8	Quenched ($3d$)
Mn^{2+}, Fe^{3+} ($3d^5$)	$^6S_{5/2}$	5.92	5.92	5.9	Half-filled
Fe^{2+} ($3d^6$)	$^5D_4$	6.70	4.90	5.4	Quenched (partial)

5.5 Van Vleck Temperature-Independent Paramagnetism [Optional / Advanced Read]

Optional Enrichment / Beyond-Syllabus Read: Van Vleck Paramagnetism

This derivation applies second-order quantum perturbation theory to explain small, temperature-independent paramagnetism in non-magnetic $J = 0$ ground states (such as Eu^{3+}). It is an optional enrichment topic that goes beyond the compulsory PH7302 semester exam syllabus.

Consider an atom or molecule whose ground state $|0\rangle$ is non-degenerate and has zero magnetic moment ($\langle 0|\mu_z|0\rangle = 0$, e.g. the $J = 0$ ground state $^7F_0$ of Eu^{3+}). In a magnetic field the perturbation $\mathcal{H}' = -\mu_z B$ mixes excited states $|s\rangle$ with $E_s - E_0 = \Delta_s \gg k_B T$ into the ground state via second-order perturbation theory:

$$\Delta E_0^{(2)} = -\sum_{s\neq 0} \frac{|\langle s|\mu_z|0\rangle|^2 B^2}{E_s - E_0} \implies \chi_{VV} = -\frac{\mu_0 N}{B}\frac{\partial \Delta E_0^{(2)}}{\partial B} = 2\mu_0 N \sum_{s\neq 0} \frac{|\langle s|\mu_z|0\rangle|^2}{E_s - E_0} > 0 \tag{33}$$

This is a small, positive, **temperature-independent** paramagnetism.

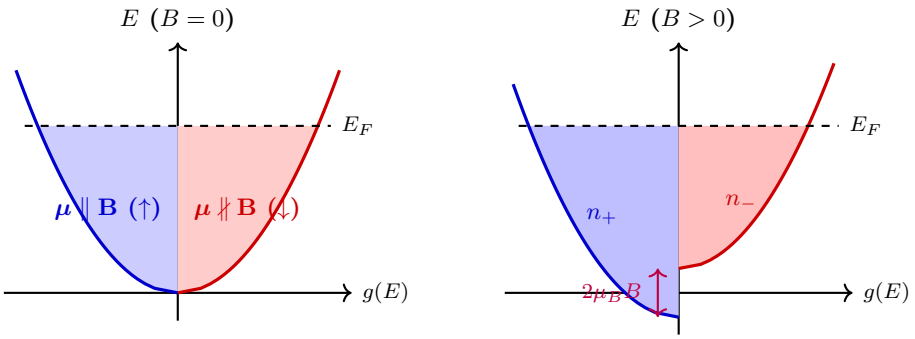
Study Takeaway: The complete map of T-dependence

$\chi = C/T$ (Curie) $\to \chi = C/(T-\Theta)$ (Curie–Weiss) $\to \chi \approx$ const (Pauli, Van Vleck) $\to T_{ord}$ with a kink or cusp. Every magnetic material in this module sits somewhere on that ladder, and the T-dependence is what identifies the mechanism.

6 Paramagnetism of Conduction Electrons: Pauli Spin Susceptibility

6.1 The Classical Puzzle and the Fermi–Dirac Resolution

If conduction electrons in a metal behaved as free localised spins they would obey Curie's law $\chi \sim N\mu_B^2/k_B T$, which diverges as $T \to 0$. Experimentally, normal metals show a weak, **temperature-independent** paramagnetism ($\sim 10^{-6}$). Wolfgang Pauli (1927) resolved this with Fermi–Dirac statistics: because of the Pauli exclusion principle, only electrons within a narrow thermal shell of order $\mu_B B$ around the Fermi energy E_F can flip their spins.



(a) Unmagnetized ($n_\uparrow = n_\downarrow$) **(b) Zeeman Split ($\Delta n \approx \mu_B B g(E_F)$)**

Figure 2: Pauli spin paramagnetism of conduction electrons at $T = 0$. Applying a field shifts the spin sub-bands by $\mp\mu_B B$, so electrons within $\mu_B B$ of E_F spill into the lower-energy sub-band. The response is small precisely because only this thin shell participates.

6.2 Derivation of the Pauli Susceptibility

Let $g(E)$ be the total density of states (both spins) per unit volume, so each spin direction carries $\frac{1}{2}g(E)$. At $T = 0$ (valid since $T \ll T_F \sim 5 \times 10^4$ K):

$$n_+ = \frac{1}{2}\int_{-\mu_B B}^{E_F} g(E + \mu_B B)\, dE \approx \frac{1}{2}\int_0^{E_F} g(E)\, dE + \frac{1}{2}\mu_B B\, g(E_F)$$
$$n_- = \frac{1}{2}\int_{+\mu_B B}^{E_F} g(E - \mu_B B)\, dE \approx \frac{1}{2}\int_0^{E_F} g(E)\, dE - \frac{1}{2}\mu_B B\, g(E_F) \tag{34}$$

The net excess of aligned moments is $\Delta n = n_+ - n_- = \mu_B B\, g(E_F)$, so

$$M = \mu_B(n_+ - n_-) = \mu_B^2 g(E_F)\, B \tag{35}$$

For a 3D free-electron gas $g(E_F) = \frac{3n}{2E_F} = \frac{3n}{2k_B T_F}$, giving the **Pauli spin susceptibility**

$$\boxed{\chi_{\text{Pauli}} = \mu_0 \mu_B^2 g(E_F) = \frac{3\mu_0 n \mu_B^2}{2k_B T_F} = \chi_{\text{Curie}} \cdot \frac{3T}{2T_F}} \tag{36}$$

Note that T appears only in the ratio T/T_F: since $T \ll T_F$, the susceptibility is essentially temperature independent — the Pauli analogue of the Van Vleck result.

- **Landau diamagnetism.** In a magnetic field the orbital motion of conduction electrons is quantised into Landau levels, producing a diamagnetic response equal to exactly one-third of the Pauli spin susceptibility (for free electrons with effective mass m^*):

$$\chi_{\text{Landau}} = -\frac{1}{3}\left(\frac{m_e}{m^*}\right)^2 \chi_{\text{Pauli}} \implies \chi_{\text{total}} = \chi_{\text{Pauli}} + \chi_{\text{Landau}} = \frac{2}{3}\chi_{\text{Pauli}} \quad (m^* = m_e) \tag{37}$$

Study Takeaway: Why Pauli paramagnetism is so weak

The Pauli susceptibility is suppressed by a factor of order $T/T_F \sim 10^{-3}$ relative to a Curie spin system of the same moment density. Physically: only the electrons in a shell of width $\mu_B B$ around E_F can respond, whereas in a Curie paramagnet *every* moment responds. This is the same Pauli-exclusion argument that produces the electronic specific heat $\propto T$ — the same tiny set of electrons does both jobs.

6.3 Cooling by Adiabatic (Isentropic) Demagnetisation [Optional / Advanced Read]

Optional Enrichment / Beyond-Syllabus Read: Cryogenic Application

Adiabatic demagnetisation demonstrates how the thermodynamic property $S = S(B/T)$ enables magnetic refrigeration into the millikelvin regime. It is an interesting experimental extension beyond the basic electronic theory syllabus.

For non-interacting spins the magnetic entropy $S(T, B)$ depends only on the population distribution, which is a function solely of $x = \mu_B B/k_B T$, i.e. $S = S(B/T)$.

1. **Step 1 — isothermal magnetisation at T_i.** Apply field B_i while in contact with a helium bath ($T_i \sim 1$ K). Spins align, the entropy drops from $S(0)$ to $S(B_i/T_i)$, and heat $\Delta Q = T_i \Delta S$ is rejected to the bath.
2. **Step 2 — adiabatic demagnetisation.** Thermally isolate the salt ($\Delta S = 0$) and slowly reduce $B_i \to B_{\text{local}}$, where $B_{\text{local}} \sim 10^{-2}$ T is the internal dipolar field. Since $S(B/T) = \text{const}$:

$$\frac{B_i}{T_i} = \frac{B_f}{T_f} \implies \boxed{T_f = T_i \frac{B_{\text{local}}}{B_i}} \tag{38}$$

Common Misconception & Pitfall: Why the cooling stops

The isentropic condition (38) assumes the spins are in equilibrium with the applied field. Once B falls to the internal dipolar field B_{local}, the spins are no longer polarised by the applied field and the salt equilibrates with its own internal field. This sets a hard floor at the millikelvin level for electronic spins. Nuclear demagnetisation (e.g. of copper) reaches the microkelvin regime because nuclear moments are $\sim 10^{-3}$ times smaller and hence the internal fields are far smaller.

7 The Exchange Interaction: Why Spins Order

7.1 Why Dipole–Dipole Coupling Cannot Explain Ferromagnetism

Conceptual Checkpoint: Order of magnitude before theory

Iron has $T_C = 1043$ K, so $k_B T_C \approx 0.090$ eV of thermal energy is needed to destroy spin alignment. Estimate the classical dipole–dipole interaction between two neighbouring Bohr magnetons separated by a lattice constant $a \approx 2$ Å. How many orders of magnitude short are you?

$$U_{\text{dip}} \approx \frac{\mu_0 \mu_B^2}{4\pi a^3} = \frac{(10^{-7}\ \text{T m/A})(9.27\times10^{-24}\ \text{J/T})^2}{(2\times10^{-10}\ \text{m})^3} \approx 1.08\times10^{-24}\ \text{J} \approx 6.7\times10^{-6}\ \text{eV} \tag{39}$$

$$\Longrightarrow \qquad T_{\text{dip}} = \frac{U_{\text{dip}}}{k_B} \approx 0.078\ \text{K} \tag{40}$$

Since $k_B T_C = 0.090$ eV, the dipolar scale is smaller by a factor of $\approx 1.3\times10^4$ — about **four orders of magnitude**. Direct magnetic dipole coupling is simply far too weak to explain room-temperature ferromagnetism.

Study Takeaway: The punchline of this section

Magnetic ordering in iron, cobalt, nickel and the rare earths is driven by the **electrostatic Coulomb interaction combined with the Pauli Exclusion Principle** — the **exchange interaction**. It is an effect of antisymmetry: it exists because electrons are fermions, and it has no classical analogue whatsoever. This is the deep reason the exchange interaction is so much stronger than the dipole interaction, even though it is "only" electrostatic.

7.2 The Two-Electron Toy Model: Heitler–London H_2

Consider two electrons at $\mathbf{r}_1, \mathbf{r}_2$ in orbitals ϕ_a, ϕ_b centred on sites a and b. The two-electron Hamiltonian $\mathcal{H} = \mathcal{H}_0 + \frac{e^2}{4\pi\epsilon_0 r_{12}}$ is *completely spin-independent*. However the total wavefunction $\Psi(\mathbf{r}_1, s_1; \mathbf{r}_2, s_2) = \psi_{\text{space}}(\mathbf{r}_1, \mathbf{r}_2)\chi_{\text{spin}}(s_1, s_2)$ must be **antisymmetric** under the simultaneous exchange $(1 \leftrightarrow 2)$.

1. **Spin singlet ($S = 0$, antiparallel spins).** $\chi_{\text{singlet}} = \frac{1}{\sqrt{2}}(\uparrow\downarrow - \downarrow\uparrow)$ is *antisymmetric*, so the spatial part must be **symmetric** (bonding: charge piles up between the nuclei):

$$\psi_S(\mathbf{r}_1, \mathbf{r}_2) = \frac{1}{\sqrt{2(1+S_{ab}^2)}}\Big[\phi_a(\mathbf{r}_1)\phi_b(\mathbf{r}_2) + \phi_b(\mathbf{r}_1)\phi_a(\mathbf{r}_2)\Big], \qquad E_S = \frac{K + J_{\text{ex}}}{1+S_{ab}^2} \approx K + J_{\text{ex}} \tag{41}$$

2. **Spin triplet ($S = 1$, parallel spins).** $\chi_{\text{triplet}} = \{\uparrow\uparrow, \frac{1}{\sqrt{2}}(\uparrow\downarrow + \downarrow\uparrow), \downarrow\downarrow\}$ is *symmetric*, so the spatial part must be **antisymmetric** (a node at $\mathbf{r}_1 = \mathbf{r}_2$, creating a **Fermi hole** that keeps parallel-spin electrons apart):

$$\psi_T(\mathbf{r}_1, \mathbf{r}_2) = \frac{1}{\sqrt{2(1-S_{ab}^2)}}\Big[\phi_a(\mathbf{r}_1)\phi_b(\mathbf{r}_2) - \phi_b(\mathbf{r}_1)\phi_a(\mathbf{r}_2)\Big], \qquad E_T = \frac{K - J_{\text{ex}}}{1-S_{ab}^2} \approx K - J_{\text{ex}} \tag{42}$$

Here K is the direct Coulomb integral and J_{ex} the **exchange integral**. The energy splitting $E_S - E_T = 2J$ depends purely on the total spin $S \in \{0, 1\}$.

Using the spin operator identity

$$\mathbf{S}_{\text{tot}}^2 = (\mathbf{s}_1 + \mathbf{s}_2)^2 = \tfrac{3}{4} + \tfrac{3}{4} + 2\,\mathbf{s}_1 \cdot \mathbf{s}_2 \implies \mathbf{s}_1 \cdot \mathbf{s}_2 = \frac{1}{2}S(S+1) - \frac{3}{4} = \begin{cases} -\frac{3}{4} & (S = 0,\ \text{singlet}) \\ +\frac{1}{4} & (S = 1,\ \text{triplet}) \end{cases} \tag{43}$$

we can write an effective spin Hamiltonian that reproduces both E_S and E_T identically:

$$\mathcal{H}_{\text{eff}} = \frac{1}{4}(E_S + 3E_T) - (E_S - E_T)\,\mathbf{s}_1 \cdot \mathbf{s}_2 = \text{const} - 2J\,\mathbf{s}_1 \cdot \mathbf{s}_2, \qquad J \equiv \frac{E_S - E_T}{2} \tag{44}$$

Generalising to a lattice of spins $\mathbf{S}_i$ gives the **Heisenberg Hamiltonian**:

$$\boxed{\mathcal{H}_{\text{Heis}} = -2\sum_{\langle i,j \rangle} J_{ij}\,\mathbf{S}_i \cdot \mathbf{S}_j \;-\; g\mu_B\,\mathbf{B} \cdot \sum_i \mathbf{S}_i} \tag{45}$$

where the sum runs over *distinct* nearest-neighbour pairs. The sign of J decides the ordering:

- $J > 0 \Rightarrow E_T < E_S \Rightarrow$ **ferromagnetic** coupling (parallel spins minimise the energy).
- $J < 0 \Rightarrow E_S < E_T \Rightarrow$ **antiferromagnetic** coupling (antiparallel spins minimise the energy, as in the covalent H_2 bond).

Common Misconception & Pitfall: Double-counting in the Heisenberg Hamiltonian

The prefactor convention matters and is a frequent source of a factor of two. In (45) the sum is over *unordered* nearest-neighbour pairs $\langle i, j \rangle$ and the prefactor is 2. If instead you write $-\sum_{i \neq j} J_{ij}\,\mathbf{S}_i \cdot \mathbf{S}_j$ (an *ordered* double sum over $i \neq j$, in which each bond appears twice), the prefactor must be 1 — or equivalently you must halve the definition of J. Every formula that follows in Chapters 6–8 uses the $\langle i, j \rangle$ convention with prefactor 2, including $T_C = 2zJS(S+1)/3k_B$.

7.3 Types of Exchange Mechanism in Solids

1. **Direct exchange (Bethe–Slater curve).** Direct overlap of neighbouring d orbitals. The sign depends critically on the ratio r_a/r_{3d} (interatomic distance divided by the radius of the $3d$ shell). Small r_a/r_{3d} (Mn, Cr) gives strong overlap → covalent bonding → $J < 0$ (antiferromagnetic). Intermediate r_a/r_{3d} (Fe, Co, Ni) gives weak overlap, where the intra-atomic Fermi-hole reduction of the Coulomb repulsion wins → $J > 0$ (ferromagnetic).

2. **Superexchange.** Indirect coupling of transition-metal cations across an intermediate non-magnetic anion (e.g. Mn^{2+}–O^{2-}–Mn^{2+} in MnO). Usually gives strong antiferromagnetic coupling ($J < 0$).

3. **RKKY indirect exchange (Ruderman–Kittel–Kasuya–Yosida).** [Optional / Advanced Read] In rare-earth metals (Gd, Tb, Dy) the localised $4f$ spins polarise the surrounding sea of conduction electrons with an oscillatory spatial decay:

$$J_{\text{RKKY}}(r) \propto \frac{\cos(2k_F r)}{(2k_F r)^3} \tag{46}$$

which can produce ferromagnetic, antiferromagnetic or helical spin order depending on the interatomic spacing.

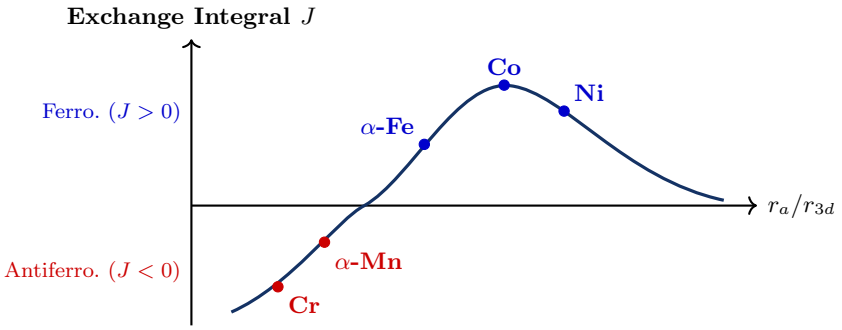


Figure 3: The Bethe–Slater curve: the sign and magnitude of the direct exchange integral J as a function of the ratio r_a/r_{3d} (interatomic distance divided by the radius of the $3d$ shell) for $3d$ transition metals. Chromium and α-Mn sit in the antiferromagnetic region; Fe, Co and Ni sit in the ferromagnetic region.

8 Ferromagnetism and Weiss Molecular Field Theory

8.1 From the Heisenberg Hamiltonian to the Weiss Exchange Field

Consider site i in (45), interacting with z nearest neighbours through J:

$$\mathcal{H}_i = -2J\,\mathbf{S}_i \cdot \sum_{\delta=1}^{z} \mathbf{S}_{i+\delta} - g\mu_B\,\mathbf{S}_i \cdot \mathbf{B}_a \tag{47}$$

In the **mean-field (Weiss molecular field) approximation** we neglect correlated fluctuations $(\mathbf{S}_i - \langle\mathbf{S}\rangle)\cdot(\mathbf{S}_j - \langle\mathbf{S}\rangle)$ and replace the neighbour spins by their thermal average $\langle\mathbf{S}_{i+\delta}\rangle = \langle\mathbf{S}\rangle$. With $\mathbf{M} = -Ng\mu_B\langle\mathbf{S}\rangle$:

$$\mathcal{H}_i^{\mathrm{MF}} = -g\mu_B\,\mathbf{S}_i\cdot(\mathbf{B}_a + \mathbf{B}_E) \implies \mathbf{B}_E = -\frac{2zJ}{g\mu_B}\langle\mathbf{S}\rangle = \frac{2zJ}{Ng^2\mu_B^2}\,\mathbf{M} \equiv \lambda\mathbf{M} \tag{48}$$

so that the effective field experienced by every spin is

$$\boxed{\mathbf{B}_{\mathrm{eff}} = \mathbf{B}_a + \lambda\mathbf{M}, \qquad \lambda = \frac{2zJ}{Ng^2\mu_B^2}} \tag{49}$$

Worked Example: The exchange field in iron

Take $T_C \approx 1000$ K for iron. The mean-field relation $T_C = \lambda C$ (derived below) gives $\lambda = T_C/C$, and with $C = \frac{\mu_0 N g^2 \mu_B^2 S(S+1)}{3k_B}$ the exchange field at saturation works out to

$$B_E = \lambda M_s \approx \frac{2zJS}{g\mu_B} \sim 10^3\ \mathrm{T} \quad (10^7\ \mathrm{G}).$$

This is roughly 10^4 times the strongest steady laboratory field ever produced. *Interpretation:* the exchange field is not an applied field at all — it is generated spontaneously by the neighbouring spins themselves, and it is enormous. Any hand-waving picture in which "the magnetisation aligns the spins" needs this number attached to it to be credible.

8.2 Paramagnetic Regime ($T > T_C$): the Curie–Weiss Law

Above the critical temperature T_C the spontaneous magnetisation vanishes and a small applied field induces a small magnetisation. *Convention for this chapter and the next: we work in CGS,*

as Kittel does. The Curie law in the field variable is $M = \frac{C}{T}H_{\text{eff}}$ with $C = \frac{Ng^2\mu_B^2 S(S+1)}{3k_B}$ (CGS). Replacing $H \to H_a + \lambda M$:

$$MT = C(H_a + \lambda M) \implies M(T - C\lambda) = CH_a \implies \boxed{\chi = \frac{M}{H_a} = \frac{C}{T - T_C}}, \qquad T_C = \lambda C \tag{50}$$

the **Curie–Weiss law**, with the ferromagnetic Curie temperature directly related to the microscopic exchange integral:

$$\boxed{T_C = \lambda C = \left(\frac{2zJ}{Ng^2\mu_B^2}\right)\left(\frac{Ng^2\mu_B^2 S(S+1)}{3k_B}\right) = \frac{2zJS(S+1)}{3k_B}} \tag{51}$$

Problem-Solving Strategy: Recipe: the Curie–Weiss law, in CGS and in SI

- **CGS:** $\chi = M/H$ and $\chi = C/(T - T_C)$ with $T_C = \lambda C$, $\lambda = 2zJ/(Ng^2\mu_B^2)$ and $C = Ng^2\mu_B^2 S(S+1)/3k_B$.
- **SI:** $\chi = \mu_0 M/B$ and $\chi = C/(T - T_C)$ with $C = \mu_0 Ng^2\mu_B^2 S(S+1)/3k_B$; the molecular-field coefficient multiplies **M** in *tesla*, $\lambda = 2zJ/(Ng^2\mu_B^2)$, so $T_C = \lambda C/\mu_0$.
- Both routes give the same physical result, $T_C = 2zJS(S+1)/3k_B$.
- **Sanity check:** $J > 0$ (ferromagnetic) $\Rightarrow T_C > 0$; $J < 0$ (antiferromagnetic) $\Rightarrow$ the same algebra gives a *negative* intercept, which is the origin of the "negative Weiss temperature" of antiferromagnets (Chapter 8).

8.3 Ferromagnetic Regime ($T < T_C$): Spontaneous Magnetisation

Set $\mathbf{B}_a = 0$, so $\mathbf{B}_{\text{eff}} = \lambda\mathbf{M}$. Substituting into the exact Brillouin equation:

$$M(T) = M_0\, B_S(x), \qquad M_0 = Ng\mu_B S, \qquad x = \frac{g\mu_B S(\lambda M)}{k_B T} \tag{52}$$

Expressing $\lambda = \frac{3k_B T_C}{Ng^2\mu_B^2 S(S+1)}$ and eliminating M in favour of x:

$$\boxed{\frac{M(T)}{M_0} = B_S(x) = \left(\frac{S+1}{3S}\right)\left(\frac{T}{T_C}\right)x} \tag{53}$$

For the simplest case $S = 1/2$ ($g = 2$), $B_{1/2}(x) = \tanh x$ and $\frac{S+1}{3S} = 1$, so

$$\boxed{m \equiv \frac{M(T)}{M_0} = \tanh\left(\frac{T_C}{T}\,m\right) \quad \Longleftrightarrow \quad \tanh x = \left(\frac{T}{T_C}\right)x} \tag{54}$$

Asymptotic behaviour near $T \to T_C^-$ (critical exponent $\beta = 1/2$). Near T_C, $m \ll 1$. Expanding $\tanh u \approx u - \frac{1}{3}u^3$:

$$m \approx \left(\frac{T_C}{T}\right)m - \frac{1}{3}\left(\frac{T_C}{T}\right)^3 m^3 \implies \frac{1}{3}\left(\frac{T_C}{T}\right)^3 m^2 = \frac{T_C - T}{T} \implies \boxed{\frac{M(T)}{M_0} \approx \sqrt{3}\left(1 - \frac{T}{T_C}\right)^{1/2}} \tag{55}$$

so the mean-field critical exponent is $\beta = 1/2$.

Asymptotic behaviour near $T \to 0$: mean-field failure. As $T \to 0$, $\tanh x \approx 1 - 2e^{-2x}$, so mean-field theory predicts an exponential approach $M(T)/M_0 \approx 1 - 2\exp(-2T_C/T)$. **Experiment instead shows a power law $\Delta M \propto T^{3/2}$.** The reason is that mean-field theory assumes a single spin must flip completely against the entire exchange field ($\Delta E = 2zJS$), ignoring collective long-wavelength **spin waves** — the subject of the next section.

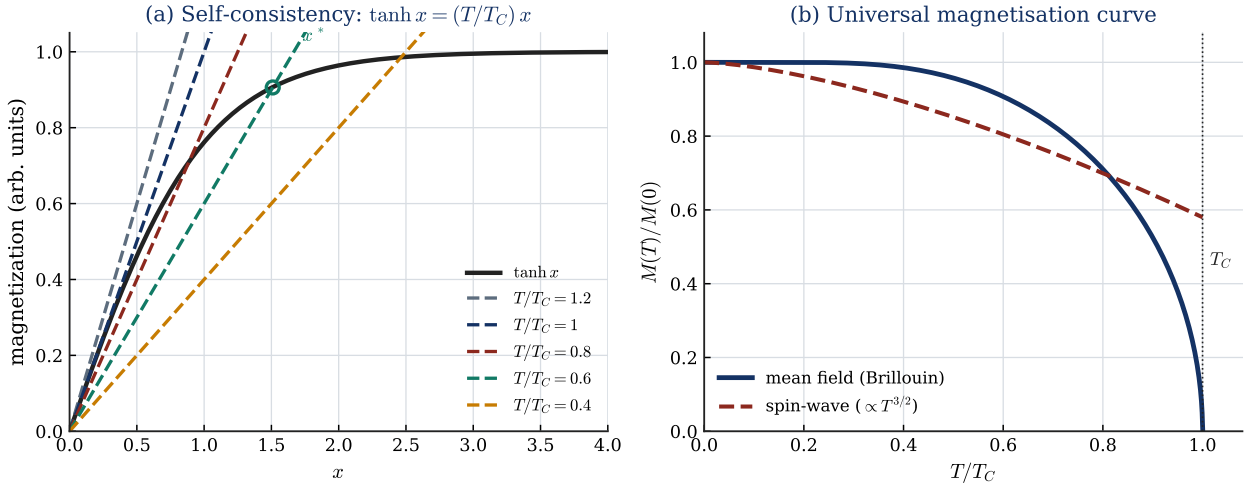


Figure 4: The mean-field self-consistency equation. **(a)** Graphical solution of $\tanh x = (T/T_C)x$: for $T > T_C$ the line is steeper than tanh and the only solution is $x = 0$; for $T < T_C$ a non-trivial solution $x^* > 0$ exists, and that solution *is* the spontaneous magnetisation. **(b)** The resulting universal magnetisation curve $M(T)/M(0)$ versus T/T_C, showing the square-root onset at T_C.

Conceptual Checkpoint: Why must the mean-field $T \to 0$ law be wrong?

Mean field gives $1 - M(T)/M_0 \propto e^{-2T_C/T}$; experiment gives $\propto T^{3/2}$. Which of the two decays *faster* as $T \to 0$, and what does that tell you about which description has more "states" available to absorb thermal energy?

8.4 Band (Itinerant) Ferromagnetism and the Stoner Criterion [Optional / Advanced Read]

Optional Enrichment / Beyond-Syllabus Read: Itinerant Magnetism

While the Heisenberg and Weiss models treat localised atomic moments, $3d$ electrons in metals (Fe, Co, Ni) are delocalised in energy bands with non-integer Bohr magneton numbers. The Stoner criterion $I \cdot g(E_F) > 1$ explains why spontaneous band splitting occurs in Fe, Co and Ni but not Cu. It is an optional enrichment topic beyond the localized Heisenberg syllabus.

In the $3d$ transition metals (Fe, Co, Ni) the $3d$ electrons are itinerant — delocalised in narrow energy bands — and carry non-integer magnetic moments per atom (Ni: 0.60 μ_B, Co: 1.72 μ_B, Fe: 2.22 μ_B). Suppose we spontaneously transfer δn electrons from the spin-down band to the spin-up band at E_F:

- **Kinetic energy cost:** $\Delta E_{\text{kin}} = +\frac{1}{2}\frac{(\delta n)^2}{g(E_F)/2} = \frac{(\delta n)^2}{g(E_F)}$
- **Exchange energy gain:** $\Delta E_{\text{ex}} = -\frac{1}{2} I (\delta n)^2$, with I the Stoner exchange parameter.

Spontaneous band splitting ($\Delta E_{\text{tot}} < 0$) occurs when the **Stoner criterion** is satisfied:

$$\boxed{I \cdot g(E_F) > 1} \tag{56}$$

This explains why Fe, Co and Ni — where E_F sits inside a high-density $3d$ peak — are ferromagnetic, whereas Cu — where E_F lies in the low-density $4s$ band above the filled $3d$ band — is not. Pd and Pt are the interesting borderline cases: "nearly ferromagnetic" Stoner-enhanced paramagnets.

9 Magnons, Domains and Hysteresis [Optional / Advanced Read]

Optional Enrichment / Beyond-Syllabus Read: Collective Excitations and Microstructure

The core PH7302 syllabus focuses on exchange and mean-field ordering. This chapter explores microscopic collective excitations (magnons / spin waves and Bloch's $T^{3/2}$ law) as well as macroscopic domain microstructures (Bloch walls, demagnetising fields and hysteresis). These sections provide essential physical depth and preparation for competitive examinations (NET/GATE), but detailed mathematical derivations here go beyond the compulsory semester syllabus.

9.1 Ferromagnetic Magnons and Bloch's $T^{3/2}$ Law [Optional / Advanced Read]

In the ground state of a ferromagnet all spins are parallel ($U_0 = -2NJS^2$ for a 1D chain). Reversing a single spin costs $\Delta E = 8JS^2$. A far lower-energy excitation is obtained by tilting all spins slightly and letting them precess on cones with a constant phase advance ka between neighbours — a **spin wave**, quantised as a boson called a **magnon**.

1. **Equations of motion.** Using

$$\frac{d\mathbf{S}_p}{dt} = \frac{1}{\hbar}\left(\mathbf{S}_p \times \text{torque}\right) = -\frac{2J}{\hbar}\mathbf{S}_p \times (\mathbf{S}_{p-1} + \mathbf{S}_{p+1}) \tag{57}$$

and linearising ($S_p^z \approx S$, $S_p^x, S_p^y \ll S$) for a plane wave $\propto e^{i(pka-\omega t)}$, one obtains the **ferromagnetic magnon dispersion**

$$\boxed{\hbar\omega_k = 4JS(1-\cos ka) \xrightarrow{ka\ll 1} 2JSa^2k^2 \equiv Dk^2 \quad (\text{3D cubic: } \hbar\omega_k = 2JSa^2k^2)} \tag{58}$$

Note the **quadratic dispersion** ($\omega \propto k^2$) at long wavelengths — the hallmark of a ferromagnet.

2. **Bloch's $T^{3/2}$ law.** Each thermally excited magnon reduces the total spin projection by one ($\Delta M = g\mu_B \sum_k n_k$), and magnons obey Bose–Einstein statistics with zero chemical potential:

$$\sum_k n_k = \int \frac{d^3k}{(2\pi)^3}\frac{1}{e^{\hbar\omega_k/k_BT}-1} = \frac{1}{2\pi^2}\int_0^\infty \frac{k^2\,dk}{e^{Dk^2/k_BT}-1} \propto \left(\frac{k_BT}{D}\right)^{3/2} \tag{59}$$

Evaluating the resulting Riemann zeta integral gives **Bloch's $T^{3/2}$ law** for both the fractional magnetisation drop and the magnon heat capacity:

$$\boxed{\frac{\Delta M(T)}{M(0)} = \frac{M(0)-M(T)}{M(0)} = \frac{0.0587}{SQ}\left(\frac{k_BT}{2JS}\right)^{3/2} \propto T^{3/2}, \qquad C_{\text{mag}} \propto T^{3/2}} \tag{60}$$

where $Q = 1, 2, 4$ for simple cubic, bcc and fcc lattices respectively.

Worked Example: Deriving the two magnon heat-capacity laws

The two cases differ through the dispersion relation, and the evaluation is carried out via standard Bose–Einstein integration.

Ferromagnet ($\hbar\omega = Dk^2$). The magnon energy per unit volume is

$$U = \frac{1}{2\pi^2}\int_0^\infty k^2\,dk\,\frac{Dk^2}{e^{Dk^2/k_BT}-1}.$$

Substituting $u = Dk^2/k_BT$, we have $k = (k_BT/D)^{1/2}u^{1/2}$, $k^2dk = \frac{1}{2}(k_BT/D)^{3/2}u^{1/2}du$, and the numerator is $Dk^2 = (k_BT)u$. The total energy density becomes

$$U = \frac{1}{4\pi^2}\left(\frac{k_BT}{D}\right)^{3/2}(k_BT)\int_0^\infty \frac{u^{3/2}}{e^u-1}du = \frac{(k_BT)^{5/2}}{4\pi^2 D^{3/2}}\,\Gamma(5/2)\zeta(5/2).$$

Using $\Gamma(5/2)\zeta(5/2) = \frac{3\sqrt{\pi}}{4}\times 1.34149 \approx 1.7833$, we find $U \approx 0.04517\,\frac{(k_BT)^{5/2}}{D^{3/2}}$. Differentiating with respect to temperature gives the heat capacity:

$$\boxed{C_{\mathrm{mag,FM}} = \frac{\partial U}{\partial T} = \frac{5}{2}\frac{k_B(k_BT)^{3/2}}{4\pi^2D^{3/2}}\Gamma(5/2)\zeta(5/2) \approx 0.1129\,k_B\left(\frac{k_BT}{D}\right)^{3/2} \propto T^{3/2}}.$$

Notice that the dimensions are completely consistent: $[k_BT/D] = \mathrm{m}^{-2}$, so $[k_B(k_BT/D)^{3/2}] = \mathrm{J/(K\cdot m^3)}$ (heat capacity per unit volume).

Antiferromagnet ($\hbar\omega = ck$, $c = 4|J|Sa$).

$$U = \frac{1}{2\pi^2}\int_0^\infty k^2\,dk\,\frac{ck}{e^{ck/k_BT}-1} = \frac{c}{2\pi^2}\left(\frac{k_BT}{c}\right)^4\int_0^\infty \frac{u^3}{e^u-1}du,$$

and since $\int_0^\infty \frac{u^3}{e^u-1}du = \frac{\pi^4}{15}$:

$$U = \frac{\pi^2}{30}\frac{(k_BT)^4}{c^3} \implies \boxed{C_{\mathrm{mag,AFM}} = \frac{2\pi^2}{15}k_B\left(\frac{k_BT}{4|J|Sa}\right)^3 \propto T^3}.$$

Rule of thumb. In d dimensions a dispersion $\omega \propto k^s$ gives $C_{\mathrm{mag}} \propto T^{d/s}$: quadratic ($s=2$) in 3D gives $T^{3/2}$; linear ($s=1$) in 3D gives T^3.

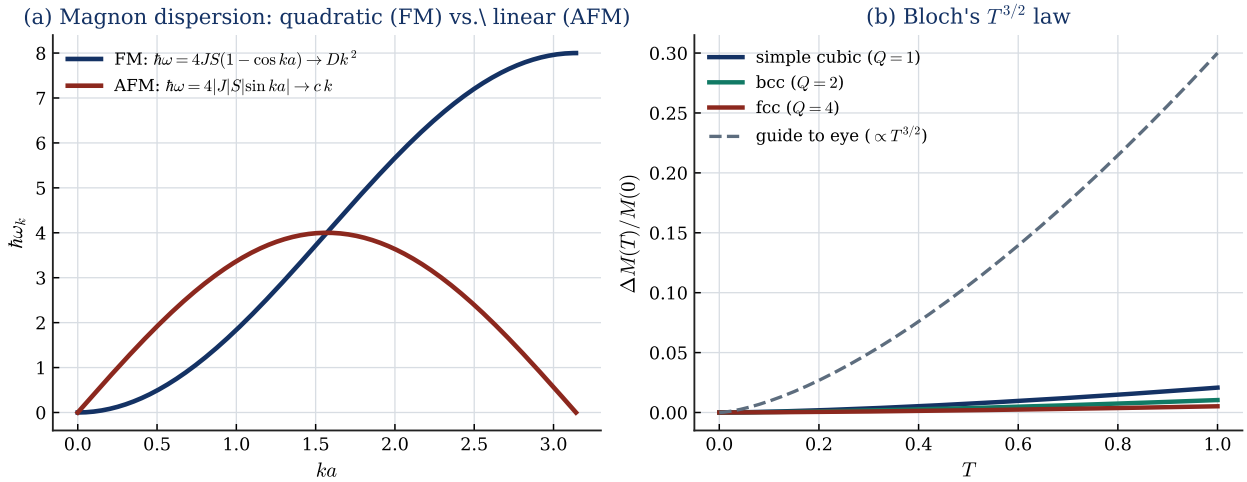


Figure 5: Magnon physics. **(a)** The ferromagnetic branch is quadratic in k and the antiferromagnetic branch is linear — the same contrast that appears in the heat-capacity exponents. **(b)** Bloch's $T^{3/2}$ law for the fractional magnetisation drop, for the three cubic lattice types ($Q = 1, 2, 4$).

9.2 Domains, Bloch Walls and Hysteresis [Optional / Advanced Read]

Why is a piece of iron at room temperature not a strong permanent magnet? Because a uniformly magnetised block creates a huge external stray field $\mathbf{B}_{\text{stray}}$ with magnetostatic energy $\int \frac{B^2}{2\mu_0}\,dV$. To minimise this energy the crystal subdivides into macroscopic **Weiss domains** with closure flux patterns.

Thickness and energy of a Bloch wall. The boundary between two antiparallel domains (a 180° rotation) is a **Bloch wall**. Why does the spin not reverse abruptly in a single atomic step?

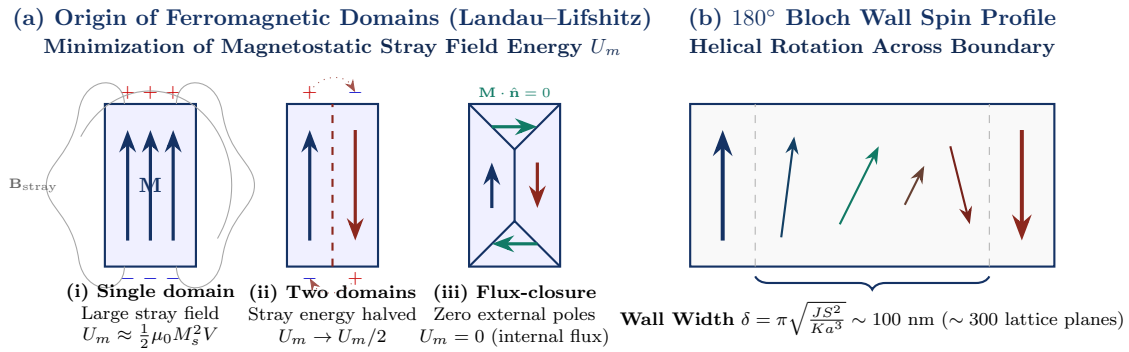
- Between two spins tilted by a small angle ϕ the exchange cost is $w_{\text{ex}} = -2JS^2\cos\phi - (-2JS^2) \approx JS^2\phi^2$.
- If a total rotation of π radians is spread over N atomic planes then $\phi = \pi/N$, and the total exchange energy of a line of N spins is

$$E_{\text{ex,line}} = N \cdot JS^2\left(\frac{\pi}{N}\right)^2 = \frac{\pi^2 JS^2}{N} \tag{61}$$

  so exchange energy alone favours $N \to \infty$ — an infinitely thick wall.
- What stops the wall from growing without limit is the **magnetocrystalline anisotropy energy** K (arising from spin–orbit coupling, which locks spins along easy crystallographic axes such as [100] in Fe or the c-axis in Co). Spins inside the wall point along hard directions, costing anisotropy energy per unit area $\sigma_{\text{anis}} \approx KNa$.
- Minimising the total wall energy per unit area $\sigma_{\text{wall}} = \sigma_{\text{ex}} + \sigma_{\text{anis}}$ with respect to N:

$$\sigma_{\text{wall}}(N) = \frac{\pi^2 JS^2}{Na^2} + KNa, \qquad \frac{\partial \sigma_{\text{wall}}}{\partial N} = 0 \tag{62}$$

$$\boxed{N = \pi\sqrt{\frac{JS^2}{Ka^3}} \sim 300 \text{ planes} \quad (\delta \sim 100 \text{ nm})} \tag{63}$$



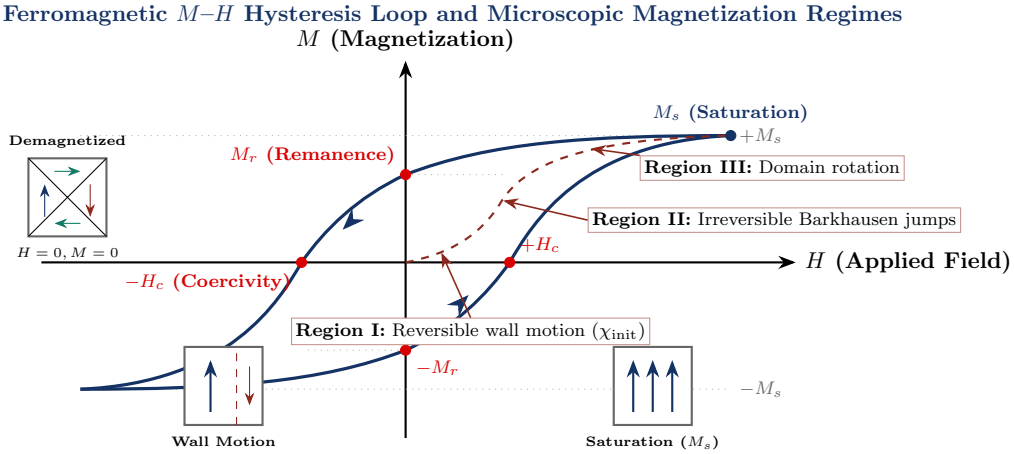


Figure 6: **(a)** The origin of ferromagnetic domains (Landau–Lifshitz energy minimization): subdivision from a single domain with strong external stray field ($U_m \approx \frac{1}{2}\mu_0 M_s^2 V$) into two antiparallel domains ($U_m/2$) and flux-closure domains with 90° closure caps where $\mathbf{M} \cdot \hat{\mathbf{n}} = 0$, completely eliminating external stray field energy ($U_m = 0$). **(b)** Internal spin structure of a 180° Bloch wall, showing gradual helical rotation across $\delta = \pi\sqrt{JS^2/Ka^3} \sim 100\,\text{nm}$ (~ 300 lattice planes) balancing exchange and anisotropy energies. **(c)** Complete ferromagnetic M–H hysteresis loop showing the virgin magnetization curve through its three physical regimes (reversible wall motion χ_{init}, irreversible Barkhausen jumps past pinning defects, and domain rotation against anisotropy), saturation M_s, remanence M_r, and coercive field H_c, accompanied by microscopic domain state insets.

10 Antiferromagnetism and Ferrimagnetism

10.1 The Two-Sublattice Néel Model

In an **antiferromagnet** ($J < 0$; e.g. MnO, FeO, CoO, NiO, MnF_2, or the itinerant spin-density-wave metal chromium) a bipartite lattice splits into two interpenetrating sublattices A (spins up, magnetisation $\mathbf{M}_A$) and B (spins down, magnetisation $\mathbf{M}_B$) with $|\mathbf{M}_A| = |\mathbf{M}_B|$ below the **Néel temperature** T_N. The net spontaneous magnetisation is therefore zero:

$$\mathbf{M}_{\text{tot}} = \mathbf{M}_A + \mathbf{M}_B = \mathbf{0} \tag{64}$$

How to Read This Figure: Experimental detection: the Shull–Smart neutron diffraction experiment (1949)

In MnO ($T_N = 118$ K) X-ray diffraction shows a chemical FCC unit cell of $a = 4.43$ Å at both 293 K and 80 K — the chemical lattice is *unchanged* by magnetic ordering. Neutron diffraction, which feels the magnetic moments, at 80 K however reveals extra half-integer superlattice Bragg peaks at $(1/2, 1/2, 1/2)$, corresponding to a magnetic unit cell of $2a = 8.85$ Å. *Why the two experiments differ:* X-rays scatter from the electron density, which does not know about spin order; neutrons carry a magnetic moment and scatter additionally from the ordered moments.

Derivation of the susceptibility above T_N. Assuming dominant antiferromagnetic coupling between the A and B sublattices (and neglecting intra-sublattice coupling), the molecular fields are

$$\mathbf{B}_A = \mathbf{B}_a - \mu\mathbf{M}_B, \qquad \mathbf{B}_B = \mathbf{B}_a - \mu\mathbf{M}_A \qquad (\mu > 0) \tag{65}$$

In the paramagnetic phase ($T > T_N$) each sublattice has Curie constant C (so $C_{\text{total}} = 2C$):

$$M_A T = C(B_a - \mu M_B), \qquad M_B T = C(B_a - \mu M_A) \tag{66}$$

1. **Néel temperature T_N.** At $B_a = 0$ a non-trivial staggered solution $M_A = -M_B \neq 0$ requires
$$\det \begin{pmatrix} T & \mu C \\ \mu C & T \end{pmatrix} = 0 \implies T^2 - (\mu C)^2 = 0 \implies \boxed{T_N = \mu C} \tag{67}$$

2. **Paramagnetic susceptibility ($T > T_N$).** Adding the two equations gives $(M_A + M_B)T = 2CB_a - \mu C(M_A + M_B)$:
$$\boxed{\chi = \frac{M_A + M_B}{B_a} = \frac{2C}{T + \mu C} = \frac{2C}{T + T_N}} \qquad \left(\text{with } AA/BB \text{ coupling: } \chi = \frac{2C}{T + \Theta}\right) \tag{68}$$

Common Misconception & Pitfall: The negative intercept of $1/\chi$ does not mean a negative ordering temperature

Notice the **plus sign** in the denominator (68). Extrapolating $1/\chi$ versus T from high temperatures therefore intercepts the temperature axis at $T = -\Theta$ (or $-T_N$). This does *not* mean the antiferromagnet orders at a negative temperature: the extrapolation is taken outside the domain of validity of the linear (paramagnetic) branch, the susceptibility is positive at every $T > 0$, and the material orders at the *positive* temperature T_N. The negative sign records only that the ordering is antiparallel. Whenever you extract Θ from a $1/\chi$ plot, extrapolate only from the region $T \gg T_N$.

10.2 Anisotropic Susceptibility Below T_N: $\chi_\perp$ versus $\chi_\parallel$ and Powder Average

Below T_N the response of a single-crystal antiferromagnet depends strongly on the orientation of the applied field relative to the easy axis of the sublattice magnetisation.

1. **Perpendicular field ($\mathbf{B}_a \perp$ easy axis, $T = 0$).** The field tilts both $\mathbf{M}_A$ and $\mathbf{M}_B$ towards $\mathbf{B}_a$ by a small angle ϕ (so the angle between them is $\pi - 2\phi$, where $\cos(\pi - 2\phi) = -\cos 2\phi \approx -(1 - 2\phi^2)$). The energy density is

$$U = \mu \mathbf{M}_A \cdot \mathbf{M}_B - \mathbf{B}_a \cdot (\mathbf{M}_A + \mathbf{M}_B) \approx -\mu M^2 (1 - 2\phi^2) - 2B_a M\phi \tag{69}$$

   Minimising with respect to ϕ:

$$\frac{\partial U}{\partial \phi} = 4\mu M^2 \phi - 2B_a M = 0 \implies \phi = \frac{B_a}{2\mu M} \implies \boxed{\chi_\perp(0) = \frac{2M\phi}{B_a} = \frac{1}{\mu} = \frac{2C}{2T_N} = \chi(T_N)} \tag{70}$$

   Remarkably, $\chi_\perp$ is **temperature independent** from $T = 0$ up to T_N in this idealised model.

2. **Parallel field ($\mathbf{B}_a \parallel$ easy axis, $T = 0$).** At $T = 0$ K the sublattices are completely saturated along $\pm\hat{z}$. A small parallel field lowers the energy of A by $-MB_a$ and raises that of B by $+MB_a$: the net energy change, and hence the magnetisation change, is zero.

$$\boxed{\chi_\parallel(0) = 0} \tag{71}$$

   As T increases from 0 to T_N, thermal agitation allows spins to flip, so $\chi_\parallel(T)$ rises smoothly from 0 to meet $\chi_\perp$ at T_N — the characteristic **cusp** at T_N.

3. **Polycrystalline (powder) average at $T = 0$ K.** Averaging over all grain orientations ($\frac{1}{3}$ parallel, $\frac{2}{3}$ perpendicular):

$$\boxed{\chi_{\text{poly}}(0) = \frac{1}{3}\chi_\parallel(0) + \frac{2}{3}\chi_\perp(0) = \frac{2}{3}\chi(T_N)} \tag{72}$$

10.3 Universal Susceptibility Comparison: Paramagnet, Ferromagnet, Antiferromagnet, and Diamagnet

A central pillar of magnetism in condensed matter physics is contrasting the temperature dependence of $\chi(T)$ and its reciprocal $1/\chi(T)$ across the fundamental magnetic classes.

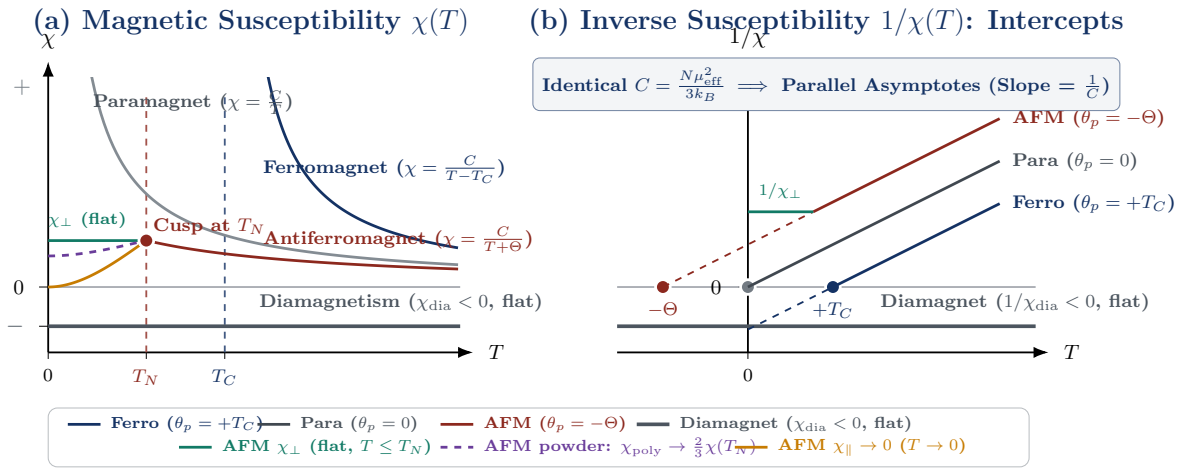


Figure 7: Universal comparison of magnetic susceptibilities across temperature. **(a)** Direct susceptibility $\chi(T)$: the ferromagnet diverges at T_C following $\chi = C/(T - T_C)$, the paramagnet decays smoothly as C/T, the antiferromagnet exhibits a characteristic cusp at T_N with three distinct low-temperature branches ($\chi_\perp$ constant, χ_{poly} dropping to $\frac{2}{3}\chi(T_N)$, and $\chi_\parallel \to 0$ at 0 K), and the diamagnet forms a temperature-independent negative baseline ($\chi_{\text{dia}} < 0$). **(b)** Inverse susceptibility $1/\chi(T)$: for materials possessing an identical density of magnetic moments (identical Curie constant C), all three paramagnetic asymptotes share the **exact same slope** $1/C$ (parallel lines). The underlying exchange coupling manifests purely as a horizontal shift along the temperature axis, yielding intercepts at $+T_C$ for ferromagnets ($\theta_p > 0$), the origin 0 for paramagnets ($\theta_p = 0$), and $-\Theta = -T_N$ for antiferromagnets ($\theta_p < 0$).

Table 5: Master examination comparison of the magnetic responses.

Magnetic Class	Sign of χ	Behavior at $T > T_{\text{order}}$	Curie–Weiss Intercept θ_p	High-T Slope $\frac{d(1/\chi)}{dT}$
Diamagnetism	Negative ($\chi < 0$)	T-independent (Lenz's orbital law)	N/A	N/A
Paramagnetism	Positive ($\chi > 0$)	$\chi = C/T$ (Curie decay)	$\theta_p = 0$ (origin)	$1/C$
Ferromagnetism	Large positive	$\chi = C/(T - T_C)$ (Curie–Weiss)	$\theta_p = +T_C > 0$	$1/C$
Antiferromagnetism	Small positive	$\chi = C/(T + \Theta)$ (Néel cusp at T_N)	$\theta_p = -\Theta < 0$	$1/C$

Common Misconception & Pitfall: Why the high-temperature slopes of $1/\chi$ vs T are identical

In examination questions, students often mistakenly wonder whether antiferromagnets or ferromagnets have a lower or higher slope in $1/\chi(T)$ compared to paramagnets. Since $1/\chi = (T - \theta_p)/C$, the slope $d(1/\chi)/dT = 1/C$ is determined solely by the Curie constant $C = \frac{N\mu_{\text{eff}}^2}{3k_B}$. For a given concentration of identical magnetic moments, C is identical, making the three Curie–Weiss asymptotes **parallel**. The sign of the exchange interaction J enters exclusively through the horizontal intercept θ_p.

10.4 Spin-Flop Transition and Antiferromagnetic Magnons [Optional / Advanced Read]

Optional Enrichment / Beyond-Syllabus Read: Optional Read: High-Field Dynamics & Excitations in Antiferromagnets

While the zero-field anisotropic susceptibilities ($\chi_\perp, \chi_\parallel$) and the universal comparison are part of the core syllabus, the behavior under strong critical magnetic fields (spin-flop) and the quantized spin-wave excitations (antiferromagnetic magnons) represent advanced topics included here for interested students.

- **Spin-flop transition.** If $\mathbf{B}_a \parallel$ easy axis is increased to a critical field $B_{\mathrm{SF}} = \sqrt{2B_E B_A}$ (with B_A the anisotropy field and $B_E = \mu M$ the exchange field), the system abruptly flops its spins perpendicular to $\mathbf{B}_a$ to take advantage of the lower magnetic energy $-\frac{1}{2}\chi_\perp B_a^2 < -\frac{1}{2}\chi_\parallel B_a^2$.
- **Antiferromagnetic magnons.** Solving the coupled equations of motion for sublattices A and B gives a **linear dispersion** (just like acoustic phonons!):

$$\boxed{\hbar\omega_k = 4|J|S|\sin(ka)| \xrightarrow{ka\ll 1} 4|J|Sa|k| \propto k} \implies C_{\mathrm{mag,AFM}} \propto T^3 \tag{73}$$

  In contrast to ferromagnetic magnons ($\omega \propto k^2 \implies C_{\mathrm{mag,FM}} \propto T^{3/2}$), antiferromagnetic magnons have a linear dispersion $\omega \propto k$ in three dimensions, yielding a T^3 magnetic heat capacity contribution.

10.5 Ferrimagnetism: Magnetite and the Ferrites [Optional / Advanced Read]

Optional Enrichment / Beyond-Syllabus Read: Ferrimagnetic Ordering

The core curriculum explicitly prescribes ferromagnetism and antiferromagnetism. Ferrimagnetism (unequal antiparallel sublattices resulting in spontaneous net magnetization, as in magnetite Fe_3O_4 and insulating RF ferrites) represents a technologically important third ordered state. This section is provided as an optional enrichment read.

In a **ferrimagnet** (Néel, 1948) the two sublattices are coupled antiferromagnetically ($B_A = B_a - \mu M_B$, $B_B = B_a - \mu M_A$) but have **unequal magnetic moments** ($C_A \neq C_B$, $M_A \neq M_B$), so a net spontaneous magnetisation survives:

$$M_{\mathrm{net}} = |M_B - M_A| \neq 0 \tag{74}$$

- **Classic example — magnetite ($\mathbf{Fe_3O_4 = Fe^{2+}O \cdot Fe_2^{3+}O_3}$, inverse spinel).** In one formula unit there are two Fe^{3+} ions ($3d^5$, $S = 5/2$, $L = 0 \Rightarrow 5\ \mu_B$ each) and one Fe^{2+} ion ($3d^6$, $S = 2$, L quenched $\Rightarrow 4\ \mu_B$). If all spins were parallel the moment would be $5 + 5 + 4 = 14\ \mu_B$ per formula unit. Experimentally $M_{\mathrm{sat}} = 4.1\ \mu_B$!
  Why? One Fe^{3+} ($5\ \mu_B$, $\downarrow$) occupies the tetrahedral A sites, while the other Fe^{3+} ($5\ \mu_B$, $\uparrow$) and the Fe^{2+} ($4\ \mu_B$, $\uparrow$) occupy the octahedral B sites. The two Fe^{3+} moments cancel exactly ($+5 - 5 = 0$), leaving only the Fe^{2+} moment of $4\ \mu_B$.
- **Susceptibility of a ferrimagnet ($T > T_C$).** Solving $M_A T = C_A(B_a - \mu M_B)$ and $M_B T = C_B(B_a - \mu M_A)$ gives

$$\boxed{\chi = \frac{M_A + M_B}{B_a} = \frac{(C_A + C_B)T - 2\mu C_A C_B}{T^2 - T_C^2}, \qquad T_C = \mu\sqrt{C_A C_B}} \tag{75}$$

  which produces a characteristic downward-curving hyperbola in $1/\chi$ versus T near T_C. Because ferrites ($MO \cdot Fe_2O_3$) are electrical insulators they suppress eddy-current losses, which makes them indispensable in high-frequency RF and microwave applications.

Study Takeaway: The six classes at a glance

Diamagnetic (small negative, T-independent, closed shells) $\rightarrow$ **Curie paramagnetic** (positive, $\propto 1/T$, localised moments) $\rightarrow$ **Pauli paramagnetic** (small positive, T-independent, itinerant) $\rightarrow$ **Ferromagnetic** (huge positive, order at T_C) $\rightarrow$ **Antiferromagnetic** (small positive, order at T_N, negative Θ) $\rightarrow$ **Ferrimagnetic** (large positive, unequal sublattices). The full quantitative comparison is tabulated in Chapter 9.

11 Master Summary and Formula Sheet

11.1 The Six Magnetic Classes

Class of material	Sign / size of χ	Temperature dependence	Microscopic origin and examples
Diamagnet	Small < 0 ($\sim -10^{-6}$)	Independent of T	Orbital Larmor shielding of closed shells (He, Ne, H_2O, Cu core, Bi).
Curie paramagnet	Positive (10^{-4}–10^{-2})	$\chi = \dfrac{C}{T}$	Permanent localised moments ($3d$, $4f$ salts, Gd^{3+}, O_2).
Pauli paramagnet	Small > 0 (10^{-6}–10^{-5})	Independent of T	Spin-split conduction electrons near E_F in metals (Na, Al, Mg).
Van Vleck paramagnet	Small > 0	Independent of T	Second-order mixing of excited states into a $J = 0$ ground state (Eu^{3+}).
Ferromagnet	Huge > 0 (10^2–10^5)	$\chi = \dfrac{C}{T - T_C}$ $(T > T_C)$	Parallel alignment ($J > 0$); spontaneous M_s below T_C (Fe, Co, Ni, Gd, EuO).
Antiferromagnet	Small > 0 (10^{-4}–10^{-3})	$\chi = \dfrac{2C}{T + \Theta}$ $(T > T_N)$	Equal antiparallel sublattices ($J < 0$); cusp at T_N (MnO, NiO, MnF_2, Cr).
Ferrimagnet	Large > 0	Hyperbolic $1/\chi$ $(T > T_C)$	Unequal antiparallel sublattices ($M_A \neq M_B$); insulating ferrites (Fe_3O_4, YIG).

Table 6: Master comparison of the magnetic classes: sign, magnitude, temperature dependence and microscopic origin.

11.2 One-Page Formula Sheet

Topic	Relevant formulae
Definitions	$\mathbf{M} = \frac{1}{V}\sum_i \boldsymbol{\mu}_i = -\frac{1}{V}\left(\frac{\partial F}{\partial \mathbf{B}}\right)_{T,V}$; $\mathbf{B} = \mu_0(\mathbf{H}+\mathbf{M})$ (SI), $\mathbf{B} = \mathbf{H} + 4\pi\mathbf{M}$ (CGS); $\chi = \mu_0 M/B$ (SI), $\chi = M/B$ (CGS), $\chi_{\mathrm{SI}} = 4\pi\chi_{\mathrm{CGS}}$; $\chi_M = \chi V_m$
Diamagnetism	$\omega_L = eB/2m$; $\mu_{\mathrm{ind}} = -\frac{Ze^2B}{4m}\langle\rho^2\rangle$; $\chi_{\mathrm{dia}} = -\frac{\mu_0 N Z e^2}{6m}\langle r^2\rangle$ (SI)
Localised paramagnetism	$g_J = 1 + \frac{J(J+1)+S(S+1)-L(L+1)}{2J(J+1)}$; $p_{\mathrm{eff}} = g_J\sqrt{J(J+1)}$; $B_J(x) = \frac{2J+1}{2J}\coth\frac{2J+1}{2J}x - \frac{1}{2J}\coth\frac{x}{2J}$; $M = N g_J \mu_B J B_J(x)$; $B_{1/2} = \tanh x$; $B_J \approx \frac{J+1}{3J}x$; $\chi = C/T$, $C = \frac{\mu_0 N g_J^2 \mu_B^2 J(J+1)}{3k_B}$ (SI)
Quenching	$p_{\mathrm{eff}}(3d) \approx 2\sqrt{S(S+1)} = \sqrt{n(n+2)}$; $p_{\mathrm{eff}}(4f) = g_J\sqrt{J(J+1)}$; $\chi_{\mathrm{VV}} = 2\mu_0 N \sum_{s\neq 0}\frac{\lvert\langle s\rvert\mu_z\lvert 0\rangle\rvert^2}{E_s - E_0}$
Itinerant paramagnetism	$\chi_{\mathrm{Pauli}} = \mu_0\mu_B^2 g(E_F) = \frac{3\mu_0 n \mu_B^2}{2k_B T_F}$; $\chi_{\mathrm{Landau}} = -\frac{1}{3}(m_e/m^*)^2\chi_{\mathrm{Pauli}}$; $T_f = T_i B_{\mathrm{local}}/B_i$
Exchange	$\mathcal{H}_{\mathrm{Heis}} = -2\sum_{\langle ij\rangle} J_{ij}\mathbf{S}_i\cdot\mathbf{S}_j - g\mu_B\mathbf{B}\cdot\sum_i \mathbf{S}_i$; $J = (E_S - E_T)/2$; $J>0$ ferromagnetic, $J<0$ antiferromagnetic; $J_{\mathrm{RKKY}}(r) \propto \frac{\cos 2k_F r}{(2k_F r)^3}$
Ferromagnetism	$\mathbf{B}_{\mathrm{eff}} = \mathbf{B}_a + \lambda\mathbf{M}$, $\lambda = \frac{2zJ}{Ng^2\mu_B^2}$; $\chi = \frac{C}{T-T_C}$; $T_C = \frac{2zJS(S+1)}{3k_B}$; $m = \tanh\frac{T_C}{T}m$; $M/M_0 \approx \sqrt{3}(1-T/T_C)^{1/2}$ ($\beta = 1/2$); Stoner: $I\,g(E_F) > 1$
Magnons	$\hbar\omega_k = 4JS(1-\cos ka) \approx 2JSa^2k^2$ (FM); $\hbar\omega_k = 4\lvert J\rvert S\rvert\sin ka\rvert \approx 4\lvert J\rvert Sa\rvert k\rvert$ (AFM); $\Delta M/M(0) = \frac{0.0587}{SQ}\left(\frac{k_B T}{2JS}\right)^{3/2}$; $C_{\mathrm{mag}} \propto T^{3/2}$ (FM), $\propto T^3$ (AFM)
Domains	$\sigma_{\mathrm{wall}} = \frac{\pi^2 JS^2}{Na^2} + KNa$; $N = \pi\sqrt{\frac{JS^2}{Ka^3}}$
Antiferromagnetism	$T_N = \mu C$; $\chi = \frac{2C}{T+T_N}$; $\chi_\perp(0) = \frac{1}{\mu} = \frac{2C}{2T_N}$; $\chi_\parallel(0) = 0$; $\chi_{\mathrm{poly}}(0) = \frac{2}{3}\chi(T_N)$; $B_{\mathrm{SF}} = \sqrt{2B_E B_A}$
Ferrimagnetism	$M_{\mathrm{net}} = \lvert M_B - M_A\rvert$; $\chi = \frac{(C_A+C_B)T - 2\mu C_A C_B}{T^2 - T_C^2}$, $T_C = \mu\sqrt{C_A C_B}$

Table 7: Consolidated formula sheet for exam revision.

A Appendix A: Python Scripts for Reproducing Every Figure

The four figures in this guide are produced by `make_figures.py`, which is distributed alongside this document in the `student_copy/` folder. Running it regenerates all of them as vector PDFs:

Listing 1: Regenerating every figure (run in the document folder).

```
python3 make_figures.py        # writes figures/fig_*.pdf
```

Below are the essential numerical kernels, each of which is self-contained.

A.1 Brillouin Function and the Curie–Weiss Fit

Listing 2: Brillouin function, its weak-field limit and the Langevin limit.

```
import numpy as np

def coth(u):
    """coth(u), safe at u = 0 and for large |u|."""
    with np.errstate(divide="ignore", invalid="ignore"):
        return 1.0 / np.tanh(u)

def brillouin(x, J):
    """Brillouin function B_J(x); uses the Taylor series near x = 0 where the
    two coth terms cancel their 1/x singularities."""
    x = np.asarray(x, dtype=float)
    a = (2.0 * J + 1.0) / (2.0 * J)
    b = 1.0 / (2.0 * J)
    with np.errstate(divide="ignore", invalid="ignore"):
        out = a * coth(a * x) - b * coth(b * x)
    small = np.abs(x) < 1e-4
    xs = x[small]
    out[small] = ((a*a - b*b) * xs / 3.0
                  - (a**4 - b**4) * xs**3 / 45.0)
    return out

def langevin(x):
    """Classical limit J -> infinity:  L(x) = coth(x) - 1/x."""
    x = np.asarray(x, dtype=float)
    with np.errstate(divide="ignore", invalid="ignore"):
        out = 1.0 / np.tanh(x) - 1.0 / x
    small = np.abs(x) < 1e-4
    out[small] = x[small] / 3.0 - x[small]**3 / 45.0
    return out

if __name__ == "__main__":
    x = np.linspace(0, 6, 1000)
    for J in (0.5, 1.5, 2.5, 3.5):
        print(f"J = {J}:  B(1) = {brillouin(1.0, J):.6f}"
              f"   weak-field slope = {(J+1)/(3*J):.6f}")
    print("Langevin L(1) =", langevin(1.0))
```

Listing 3: Solving the mean-field self-consistency equation by bisection.

```
import numpy as np

def spontaneous_magnetisation(T_over_Tc, S=0.5):
    """Return m = M(T)/M_0 from  m = B_S(m T_C / T).

    For S = 1/2 this reduces to m = tanh(m T_C / T).  We solve the
    general Brillouin form by bisection on [0, 1].
    """
    t = T_over_Tc
```

```
    if t <= 0:
        return 1.0
    def BJ(x, S):
        a = (2.0 * S + 1.0) / (2.0 * S)
        b = 1.0 / (2.0 * S)
        with np.errstate(divide="ignore", invalid="ignore"):
            return a / np.tanh(a * x) - b / np.tanh(b * x)
    lo, hi = 0.0, 1.0
    for _ in range(200):
        mid = 0.5 * (lo + hi)
        # f(m) = BJ(m/t) - m ; f(0) = 0 (always a root), f(1) < 0
        if BJ(mid / t, S) > mid:
            lo = mid
        else:
            hi = mid
    return 0.5 * (lo + hi)

if __name__ == "__main__":
    for t in (0.4, 0.6, 0.8, 0.9, 0.99, 1.01):
        print(f"T/Tc = {t:.2f}   m = {spontaneous_magnetisation(t):.6f}")
    # Compare the exact solution with the asymptotic form sqrt(3)(1-t)^(1/2)
    for t in (0.9, 0.95, 0.99):
        exact = spontaneous_magnetisation(t)
        asym = np.sqrt(3.0) * np.sqrt(1.0 - t)
        print(f"t = {t}: exact {exact:.4f}  asymptotic {asym:.4f}")
```

A.2 Susceptibility Comparison and Magnon Heat Capacity

Listing 4: The three magnetic classes and their 1/chi behaviour.

```
import numpy as np

C, Tc, TN = 1.0, 1.0, 1.0

T   = np.linspace(0.02, 3.0, 1500)
Tg  = np.linspace(Tc + 0.015, 3.0, 900)      # above Tc
Ta  = np.linspace(TN + 0.015, 3.0, 900)      # above TN
Tn  = np.linspace(0.02, TN - 0.005, 900)     # below TN

chi_para  = C / T
chi_ferro = C / (Tg - Tc)
chi_afm   = 2 * C / (Ta + TN)

chi_perp  = np.full_like(Tn, 2 * C / (2 * TN))  # flat below TN
chi_par   = np.zeros_like(Tn)                   # -> 0 below TN
chi_poly  = (2.0 / 3.0) * chi_perp              # powder average

# 1/chi intercepts: paramagnet -> 0, ferromagnet -> +Tc,
#                   antiferromagnet -> -TN
if __name__ == "__main__":
    print("ferromagnet  1/chi intercept :", Tc)
    print("antiferromag. 1/chi intercept :", -TN)
    print("chi_perp(0)  =", chi_perp[0], "  chi_poly(0) =", chi_poly[0])
```

Listing 5: Magnon heat capacity: $T^{3/2}$ (ferromagnet) and T^3 (antiferromagnet).

```
import numpy as np

def C_mag_ferro(T, D, kB=1.380649e-23):
    """C = 0.1129 k_B (k_B T / D)^{3/2}   (D in J m^2)."""
    return 0.1129 * kB * (kB * T / D)**1.5

```

```
def C_mag_afm(T, c, kB=1.380649e-23):
    """C = (2 pi^2 / 15) k_B (k_B T / c)^3   (c = 4|J|Sa in J m)."""
    return (2 * np.pi**2 / 15.0) * kB * (kB * T / c)**3

def magnon_energy_fm(k, J, S, a):
    """hbar omega = 4 J S (1 - cos k a)."""
    return 4 * J * S * (1 - np.cos(k * a))

def magnon_energy_afm(k, J, S, a):
    """hbar omega = 4 |J| S |sin k a|."""
    return 4 * abs(J) * S * np.abs(np.sin(k * a))

if __name__ == "__main__":
    # Iron-like parameters
    J, S, a = 2.1e-21, 1.0, 2.87e-10
    D = 2 * J * S * a**2
    c = 4 * abs(J) * S * a
    print(f"D = {D:.3e} J m^2 ,  c = {c:.3e} J m")
    print("C_FM(100 K) =", C_mag_ferro(100.0, D), "J/K per unit volume")
    print("C_AFM(100 K)=", C_mag_afm(100.0, c),  "J/K per unit volume")
```

A.3 Numerical Utilities Used in the Assignment

Listing 6: Constants and the formulae needed for the assignment answers.

```python
import numpy as np

# --- constants (SI) ---
e    = 1.602176634e-19      # C
me   = 9.1093837015e-31     # kg
mu0  = 4e-7 * np.pi         # H/m
kB   = 1.380649e-23         # J/K
NA   = 6.02214076e23        # 1/mol
muB  = e * 1.054571817e-34 / (2 * me)   # Bohr magneton, J/T
a0   = 5.29177210903e-11    # m

def molar_diamagnetic_susceptibility(Z, r2_mean):
    """chi_M (m^3/mol) for an atom with Z electrons and <r^2> (m^2)."""
    return -mu0 * NA * Z * e**2 * r2_mean / (6 * me)

def chi_pauli(n, TF):
    """Pauli spin susceptibility (SI) for n electrons/m^3 at Fermi temp TF."""
    return 3 * mu0 * n * muB**2 / (2 * kB * TF)

def Tc_weiss(z, J, S):
    """Curie temperature from the exchange integral J (joules)."""
    return 2 * z * J * S * (S + 1) / (3 * kB)

def beisenfeld(J, z, S, g=2.0):
    """Weiss molecular field B_E (tesla) for H = -2 J sum_<ij> S_i . S_j."""
    return 2 * z * J * S / (g * muB)

def bloch_law(T, J, S, Q):
    """Bloch's T^{3/2} fractional magnetisation drop dM/M(0)."""
    return 0.0587 / (S * Q) * (kB * T / (2 * J * S))**1.5

def TN_neel(z, Jabs, S):
    """Neel temperature for an antiferromagnet (|J| in joules)."""
    return 2 * z * Jabs * S * (S + 1) / (3 * kB)

if __name__ == "__main__":
    print("Hydrogen  chi_M =", molar_diamagnetic_susceptibility(1, 3 * a0**2), "m
        ^3/mol")
    print("Na        chi_P =", chi_pauli(2.54e28, 3.7e4))
    print("Fe (S=1)  T_C  =", Tc_weiss(8, 8.43e-3 * e, 1.0), "K")
    print("Fe (S=1)  B_E  =", beisenfeld(8.43e-3 * e, 8, 1.0), "T")
```

B Appendix B: Glossary of Symbols

Symbol	Meaning
a	Lattice constant.
a_0	Bohr radius.
B, $\mathbf{B}$	Magnetic flux density (tesla); the field that charges experience.
B_a, $\mathbf{B}_a$	Applied field.
B_E, $\mathbf{B}_E$	Weiss molecular (exchange) field.
B_A	Anisotropy field.
B_{SF}	Spin-flop field.
B_C	Coercive field, $B_C = \mu_0 H_C$.
C	Curie constant.
C_A, C_B	Curie constants of sublattices A and B.
c	Magnon velocity, $c = 4\lvert J\rvert Sa$ (antiferromagnet).
D	Spin-wave stiffness constant, $\hbar\omega = Dk^2$.
E_F	Fermi energy.
E_S, E_T	Singlet and triplet energies of a two-electron (Heitler–London) state.
F	Helmholtz free energy.
g, g_J, g_0	Electron g-factor, Landé g-factor, free-electron g-factor.
H, $\mathbf{H}$	Magnetic field intensity (A/m in SI).
J, J_{ij}	Exchange integral. Positive = ferromagnetic.
k, k_F	Wavevector, Fermi wavevector.
K	Magnetocrystalline anisotropy constant.
k_B	Boltzmann constant.
L	Orbital angular momentum quantum number; also the Langevin function $L(x)$.
M, $\mathbf{M}$	Magnetisation (A/m in SI).
M_0, M_{sat}	Saturation magnetisation.
M_A, M_B	Sublattice magnetisations.
M_r, M_s	Remanence, saturation magnetisation (hysteresis loop).
$m = M/M_0$	Reduced spontaneous magnetisation.
N	Number density of magnetic moments (per unit volume).
N_A	Avogadro constant.
p_{eff}	Effective Bohr magneton number, $g_J\sqrt{J(J+1)}$.
Q	Lattice factor in Bloch's law: 1 (sc), 2 (bcc), 4 (fcc).
q	Electronic charge.
S, $\mathbf{S}$	Spin quantum number and spin operator (dimensionless).
T_C	Curie temperature (ferromagnetic).
T_N	Néel temperature (antiferromagnetic).
θ, Θ	Weiss temperature (intercept of the $1/\chi$ plot).
V, V_m	Volume, molar volume.
x	Dimensionless field, $x = g_J\mu_B JB/k_BT$.
Z, Z_1	Partition function (total, single ion).
z	Number of nearest neighbours (coordination number).
λ	Weiss molecular-field constant.
μ	Antiferromagnetic molecular-field coefficient.
μ_B	Bohr magneton.
ρ	Mass density (or, in Section 7, the electron charge density).
σ_{wall}	Domain-wall energy per unit area.
χ, χ_M	Volume and molar susceptibility.

Symbol	Meaning
$\chi_{\parallel}$, $\chi_{\perp}$	Antiferromagnet susceptibility for a field parallel/perpendicular to the easy axis.
ω_L	Larmor precession frequency.
ω_k	Magnon frequency.

C Appendix C: References and Further Reading

1. **C. Kittel**, *Introduction to Solid State Physics*, 8th ed. (Wiley, 2004), Chapters 14–16: Magnetism. The source of the molecular-field, spin-wave and domain-wall treatments used in this module.
2. **N. W. Ashcroft and N. D. Mermin**, *Solid State Physics* (Saunders College, 1976), Chapters 31–33: Magnetism. The most readable treatment of the Bohr–van Leeuwen theorem and the two-electron exchange model.
3. **A. H. Wilson**, *Theory of Metals*, 2nd ed. (Cambridge University Press, 1954). The classic treatment of itinerant-electron magnetism and of the Stoner criterion $I\,g(E_F) > 1$.
4. **G. Rangarajan**, *Principles of Condensed Matter Physics*, NPTEL course 115106061, Lectures 19–26: Magnetism. The NPTEL lecture series accompanying this module.
5. **S. Blundell**, *Magnetism in Condensed Matter* (Oxford University Press, 2007). An excellent modern treatment of exchange mechanisms, spin waves and magnetic ordering; recommended for the RKKY and superexchange discussion.
6. **B. D. Cullity and C. D. Graham**, *Introduction to Magnetic Materials*, 2nd ed. (Wiley, 2009). The standard reference for magnetic hysteresis, domains and anisotropy.
7. **H. M. Rosenberg**, *The Solid State*, 4th ed. (Oxford University Press, 2015). A concise and very readable survey of magnetism, with useful tables of susceptibilities, Curie and Néel temperatures.
8. **P. M. Chaikin and T. C. Lubensky**, *Principles of Condensed Matter Physics* (Cambridge University Press, 1995), Chapter 11: A modern graduate-level treatment of the magnetic phases.
9. **D. J. Griffiths**, *Introduction to Elementary Particles*, 2nd ed. (Harper, 2008), Chapter 11: the quantum-statistics background behind the Curie and Pauli results.

Study Takeaway: How to use the references

Kittel and Ashcroft & Mermin cover the derivations in this module at roughly the same depth; Kittel is the faster read for the mean-field and spin-wave material, Ashcroft & Mermin for the exchange mechanism. Blundell is the best single supplementary source for the modern perspective, and Cullity & Graham is the place to go for anything to do with hysteresis and permanent magnets. The NPTEL lectures by Prof. G. Rangarajan follow the same sequence as this module and are freely available.